

Data Structures and Algorithms for Problem Solving (CS1.304)

Interval Scheduling

Instructor: Kshitij Gajjar

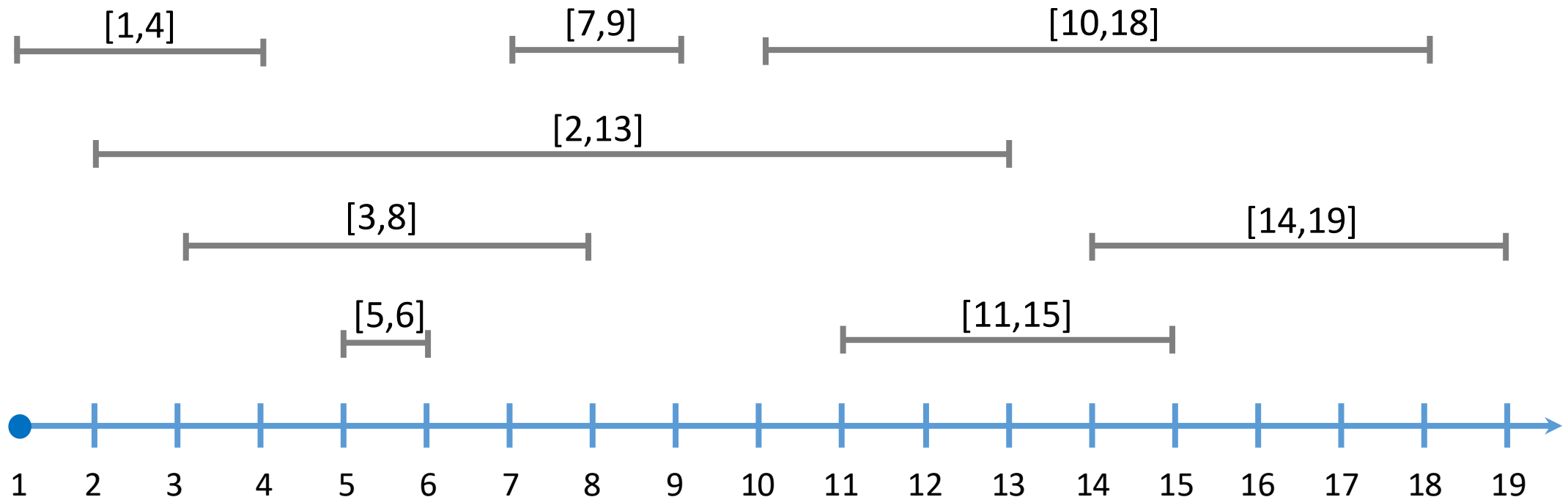
Interval Scheduling

- The start time and end time for each class is given by its instructor.
- What is the maximum number of classes that can be scheduled in one classroom in such a way that there are no clashes?

Instructor	Start time	End time	Duration
Prof. Anu	1	4	3
Prof. Ben	7	9	2
Prof. Cam	10	18	8
Prof. Dan	2	13	11
Prof. Eva	3	8	5
Prof. Fez	14	19	5
Prof. Gus	5	6	1
Prof. Hal	11	15	4

Interval Scheduling

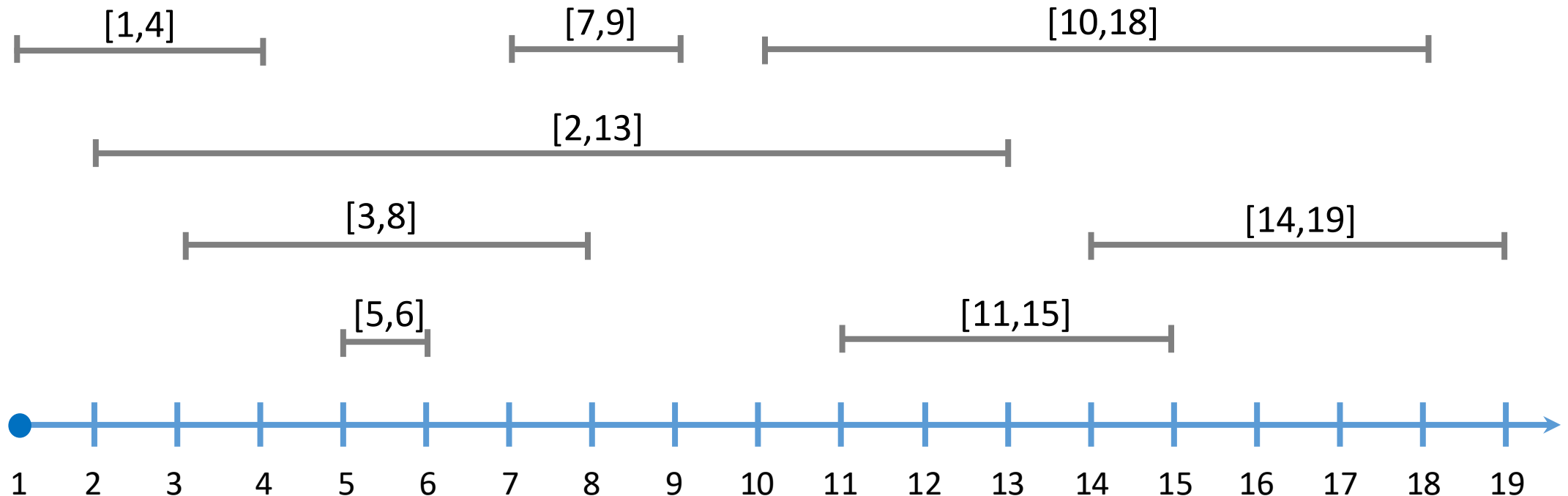
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Algorithm 1

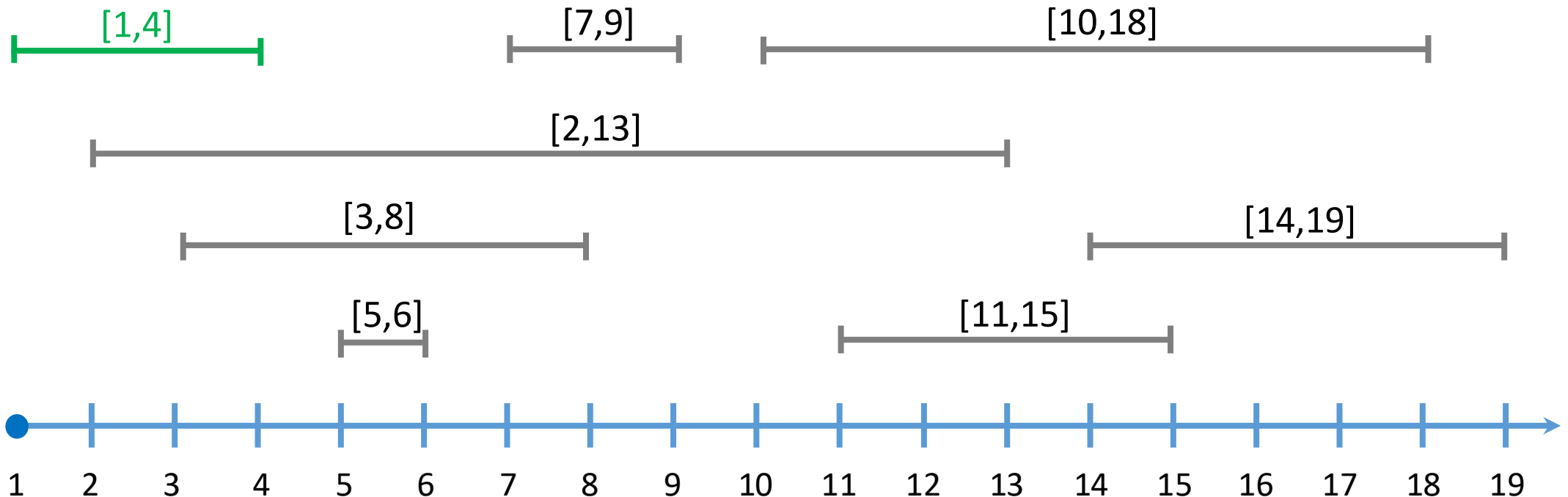
Algorithm 1

- Let us propose an algorithm: EARLIEST-START-TIME.
- From left to right, select the interval with the earliest start time.



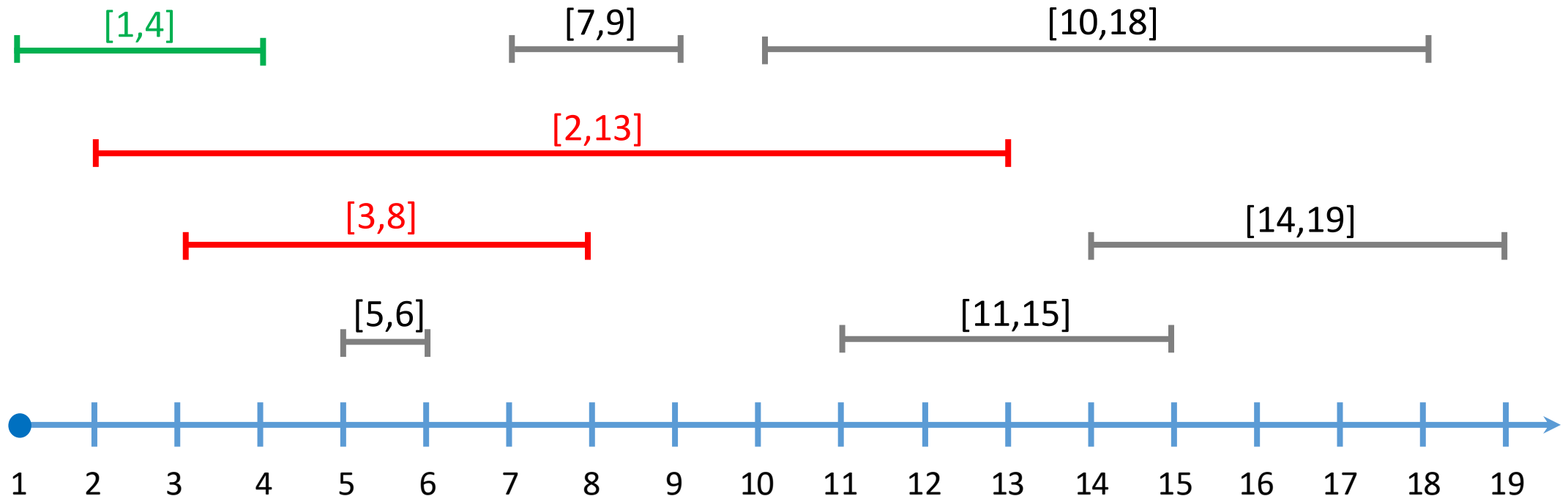
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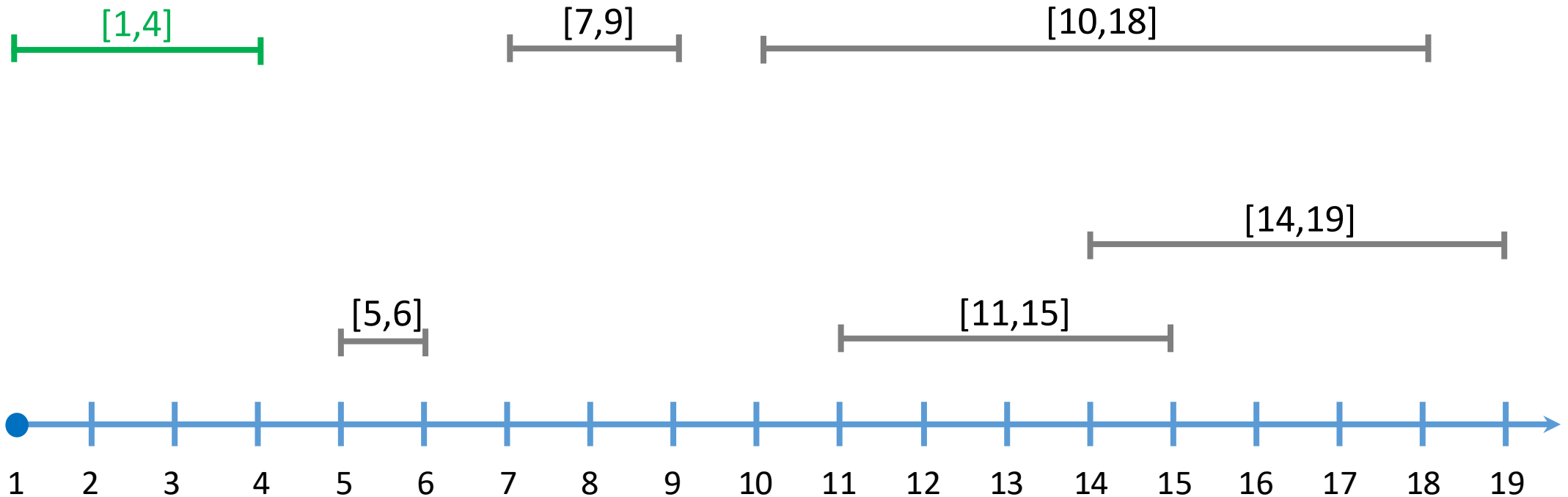
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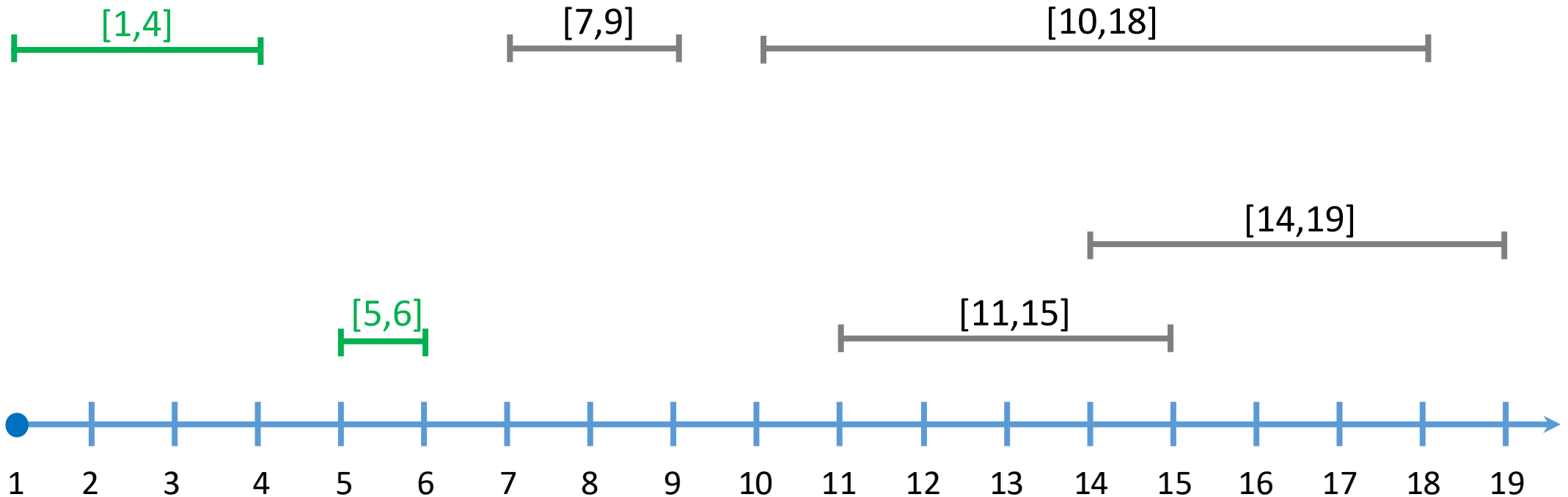
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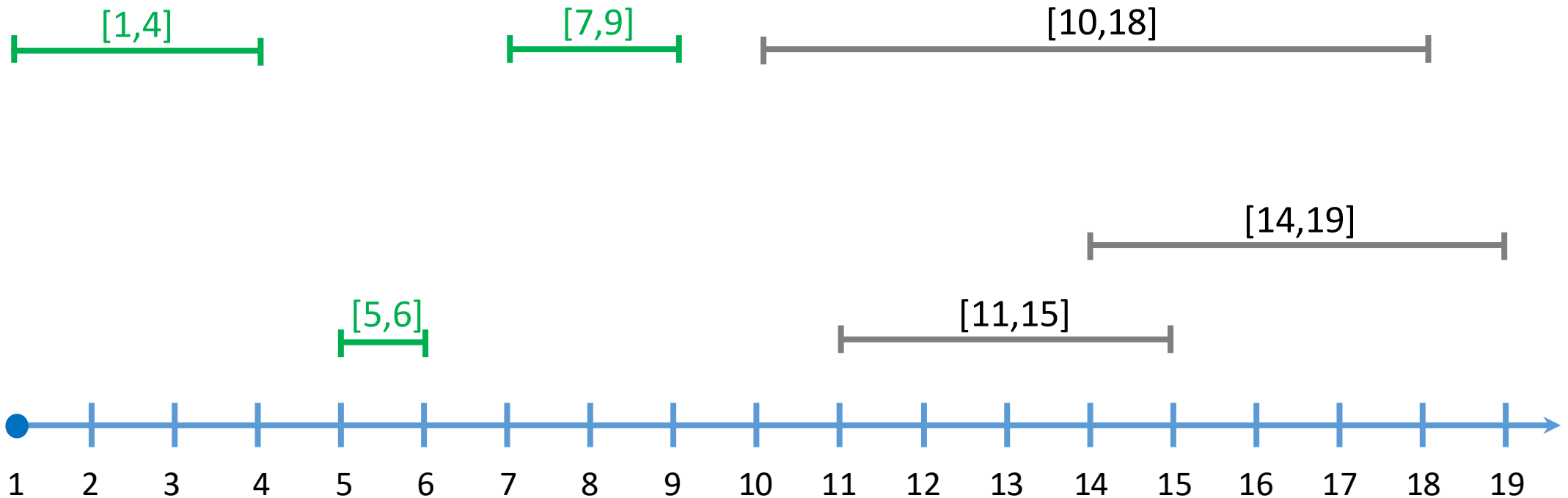
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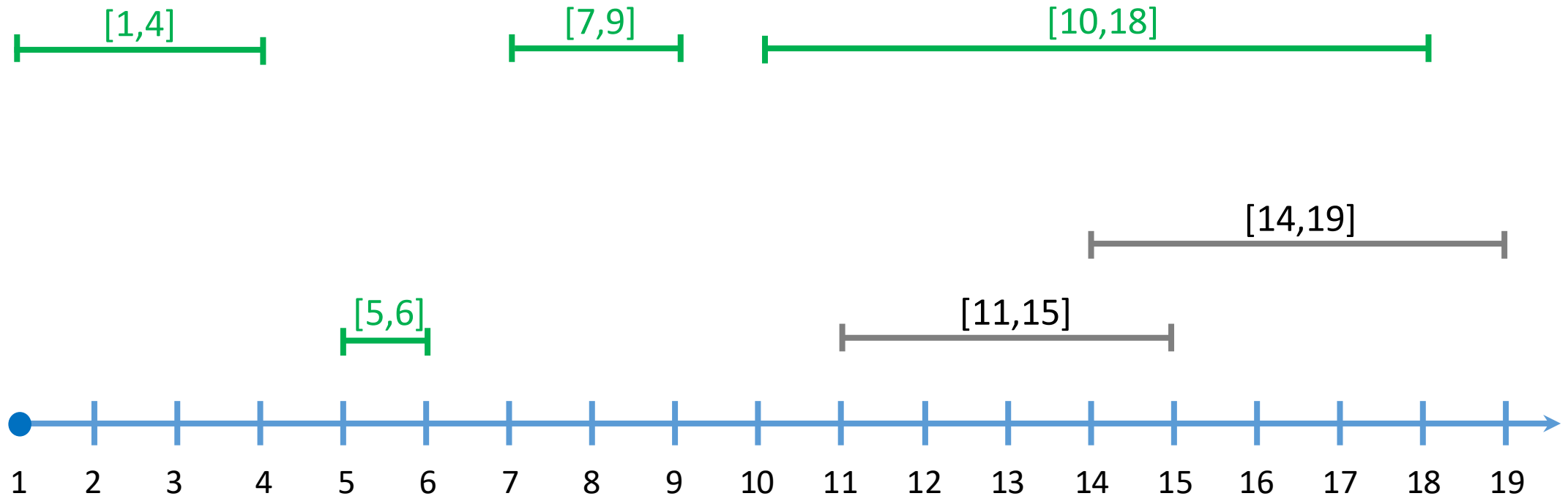
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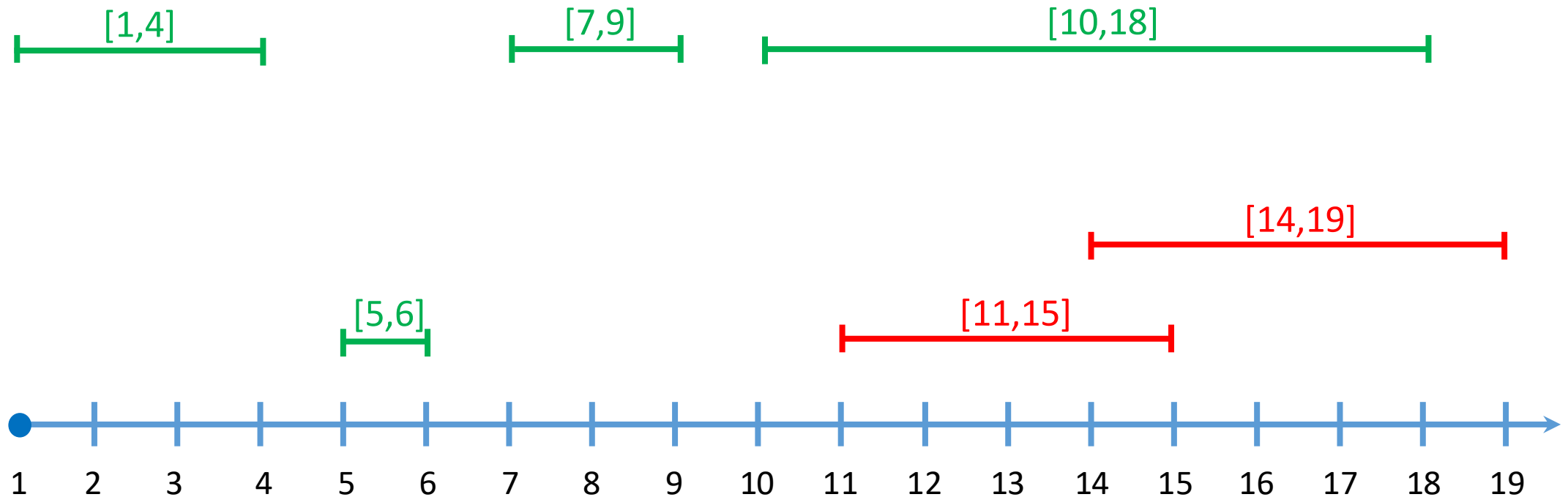
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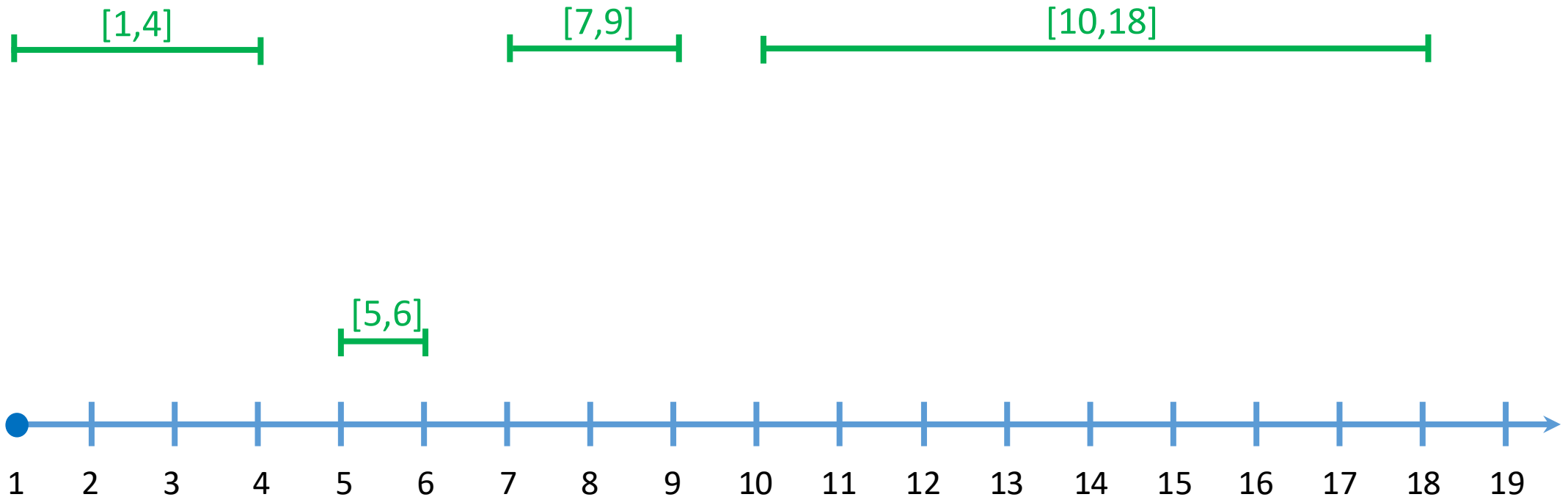
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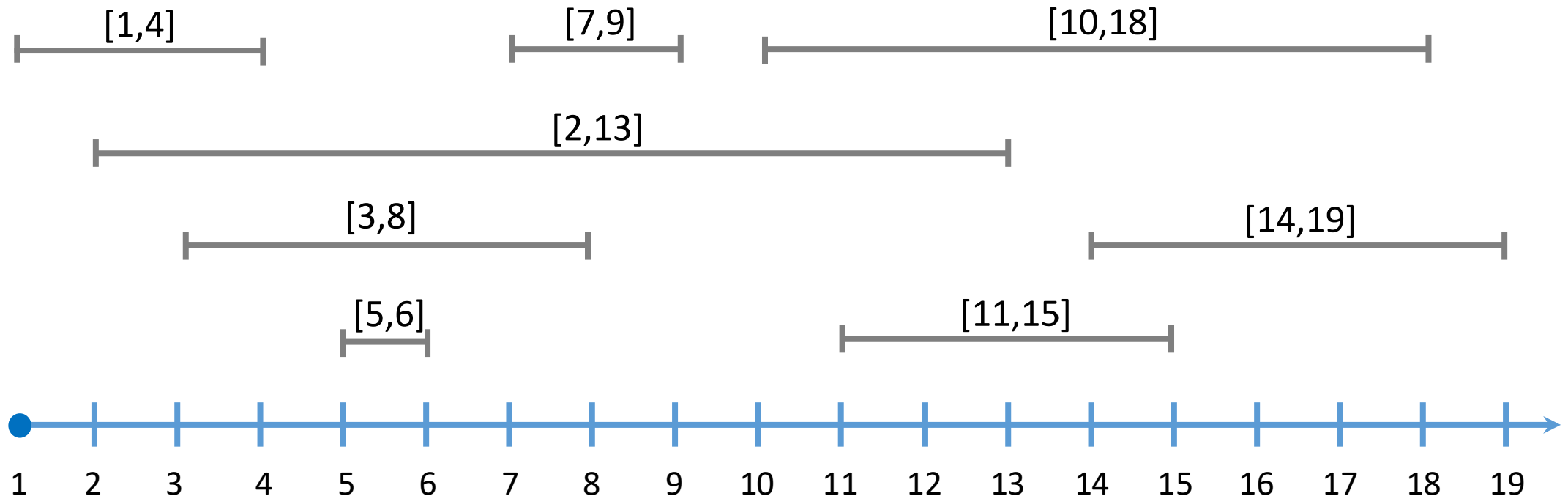
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- Does **Algorithm 1** always output a set of intervals that don't clash with each other?
- YES.
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- NO.

Algorithm 2

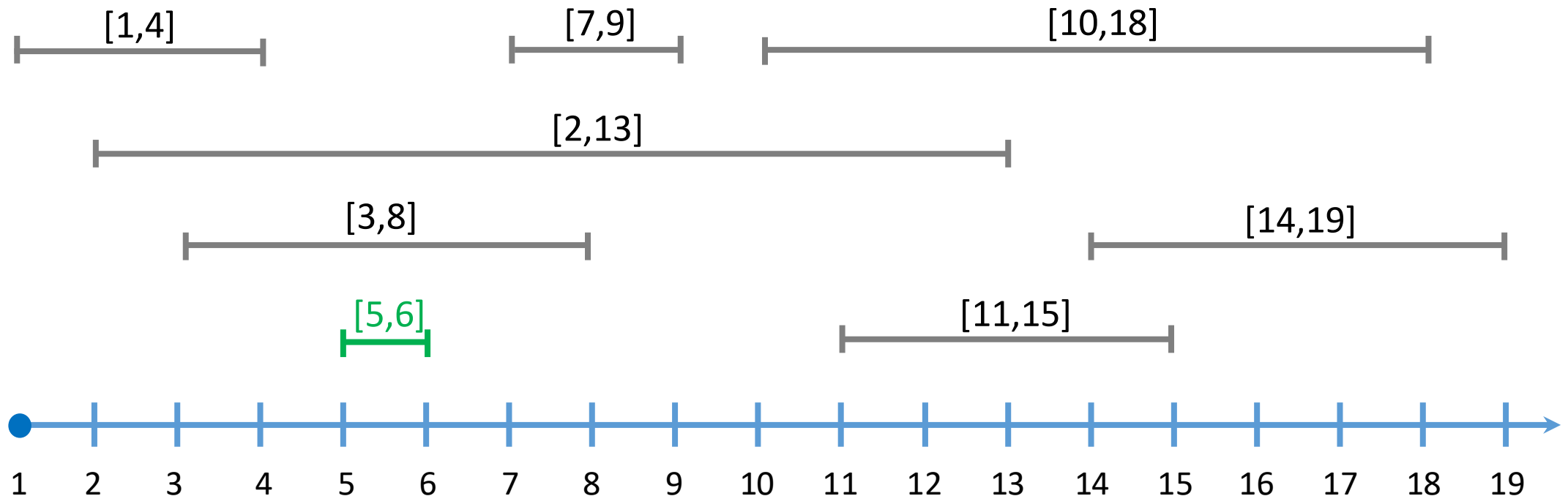
Algorithm 2

- Let us propose an algorithm: SELECT-SMALLEST-INTERVAL.
- From left to right, select the interval with the smallest time duration.



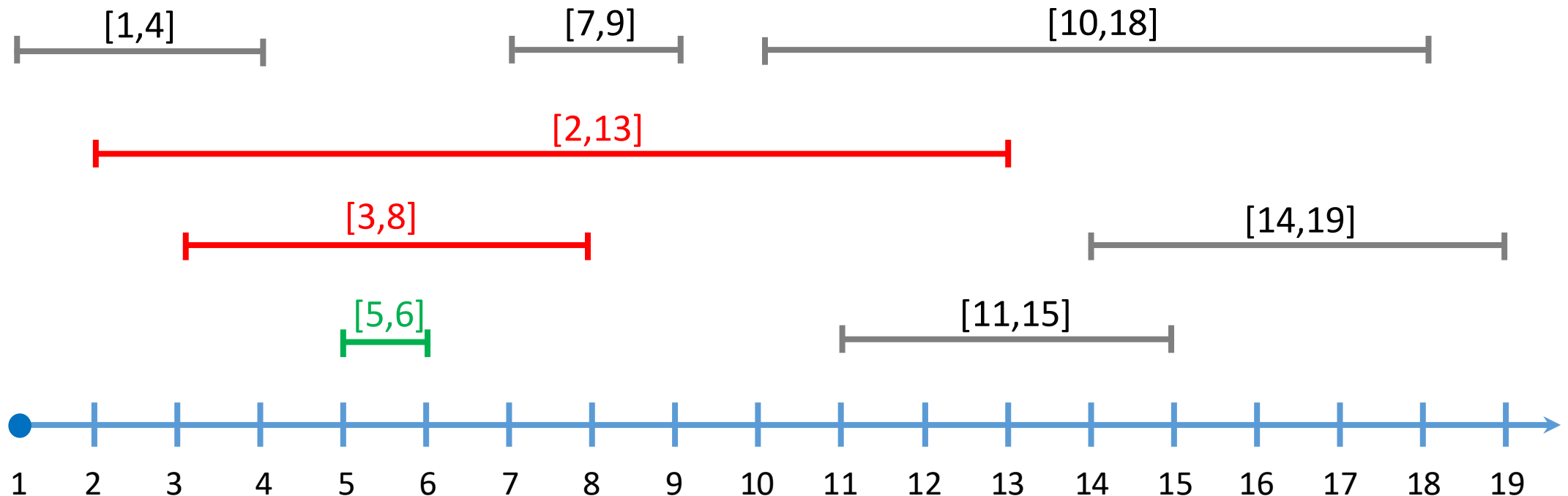
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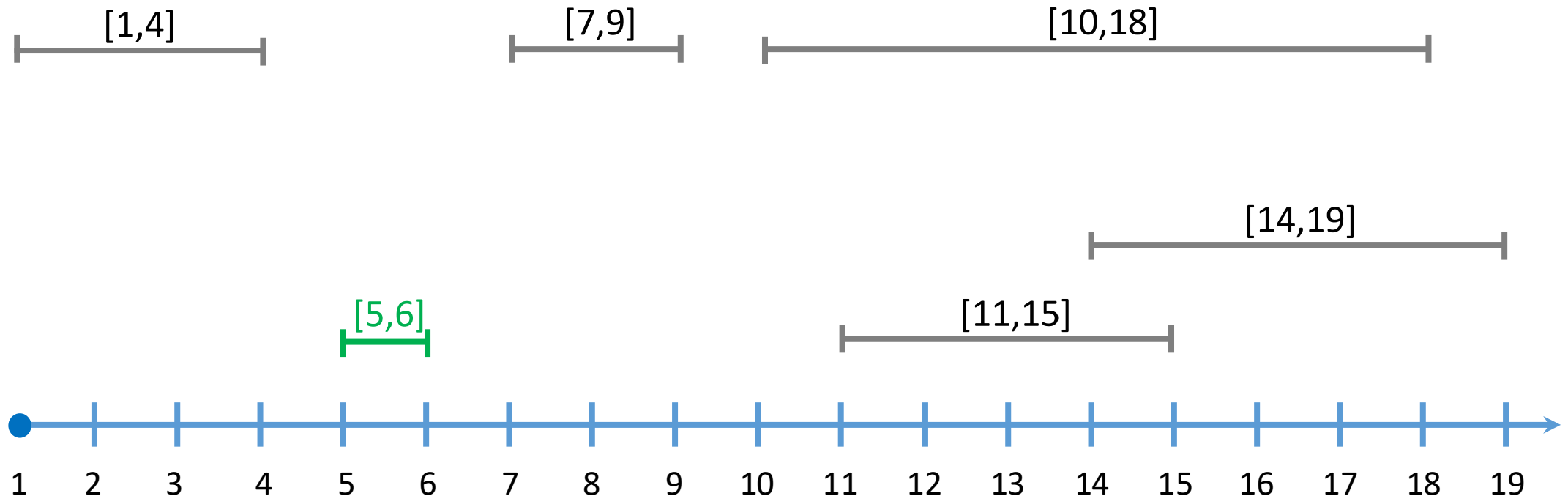
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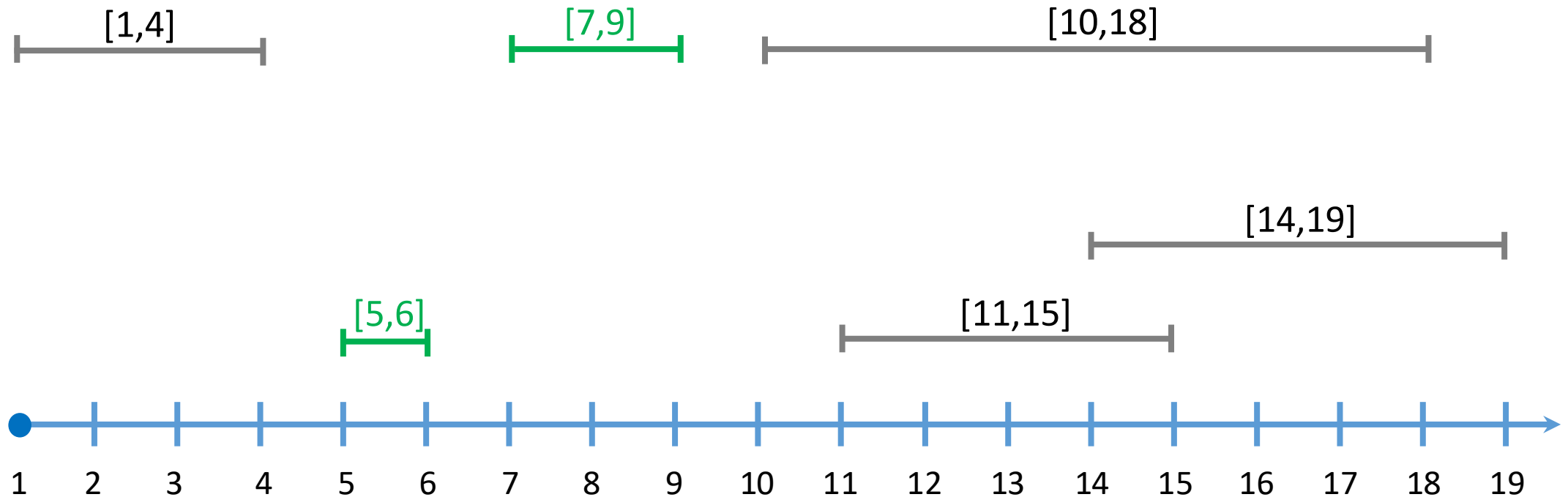
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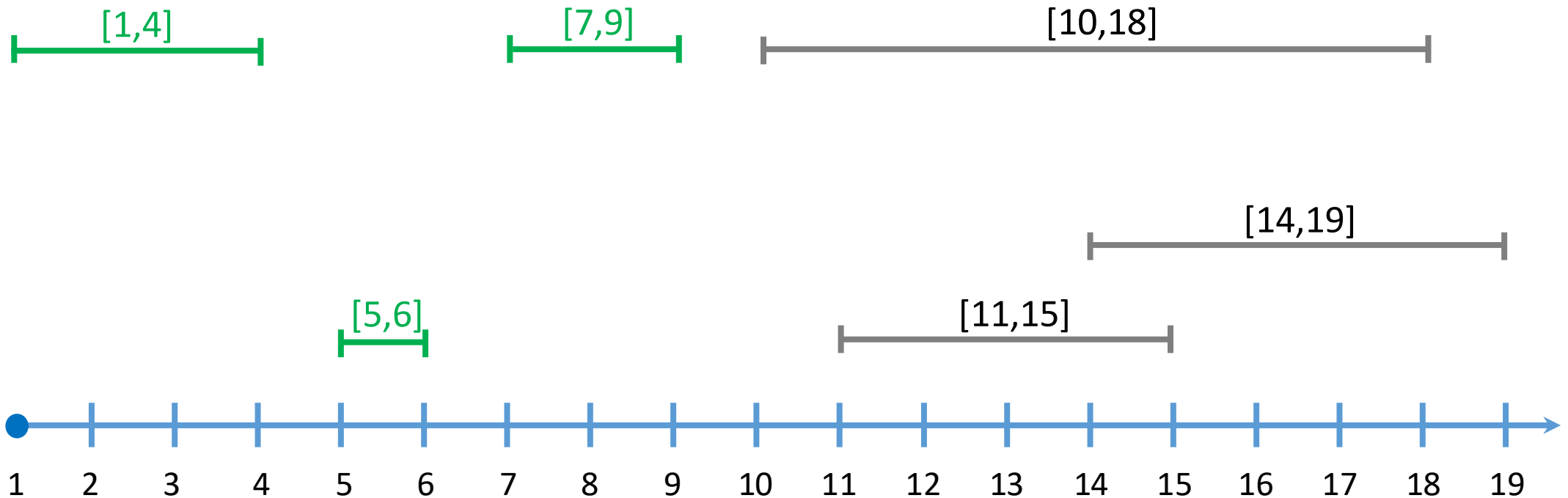
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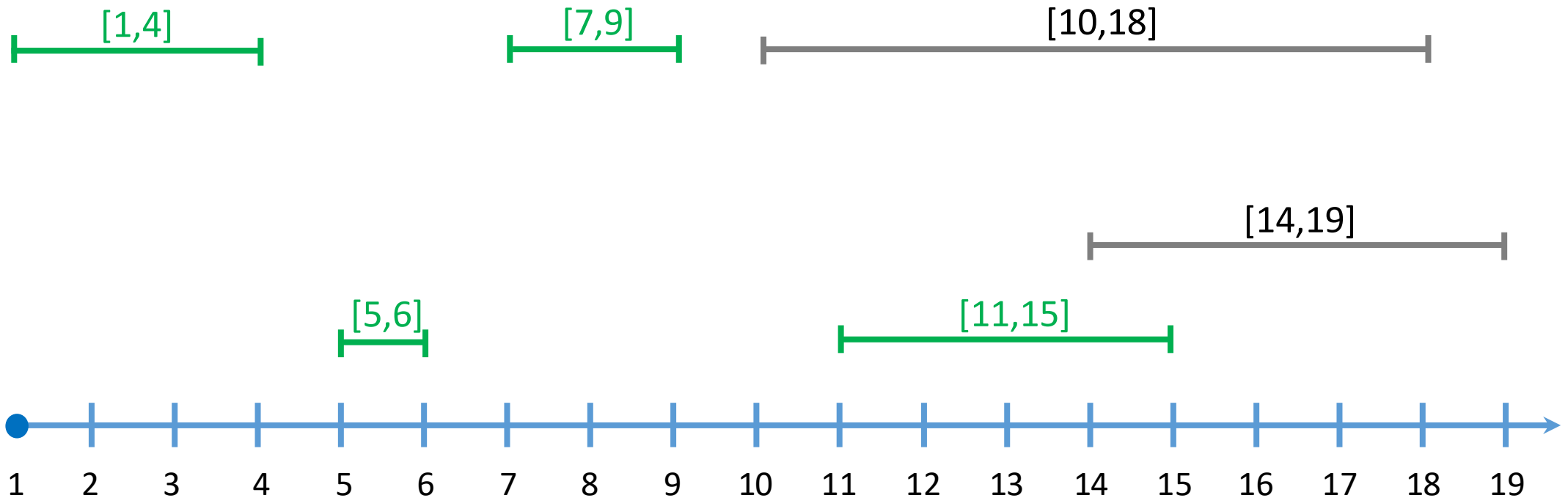
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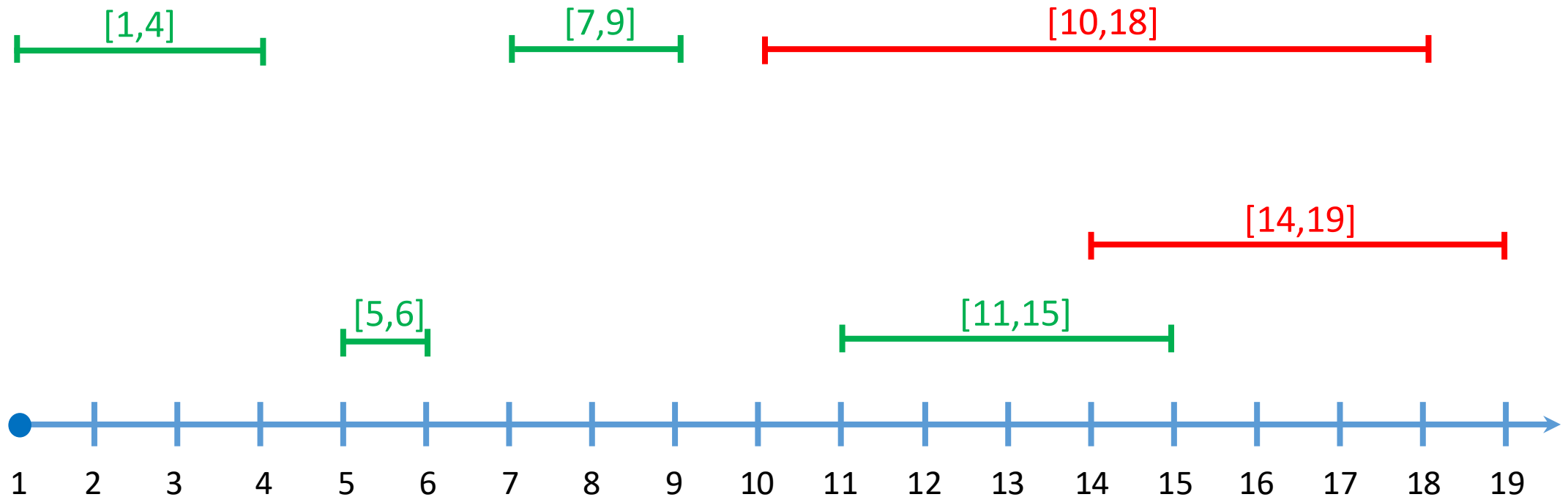
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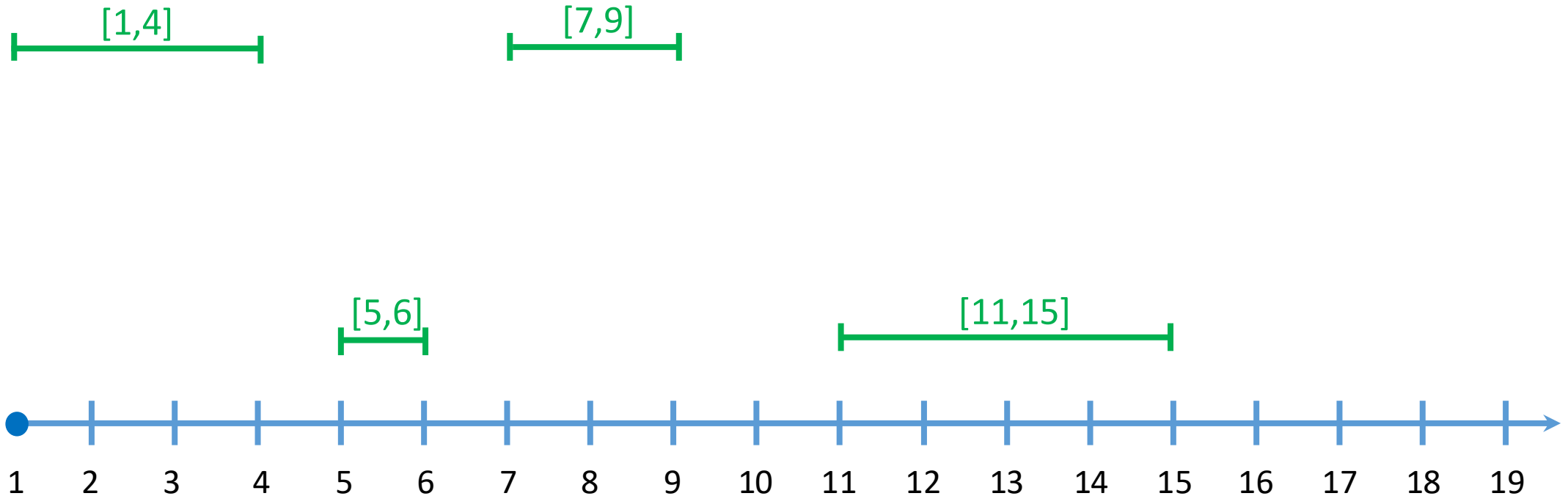
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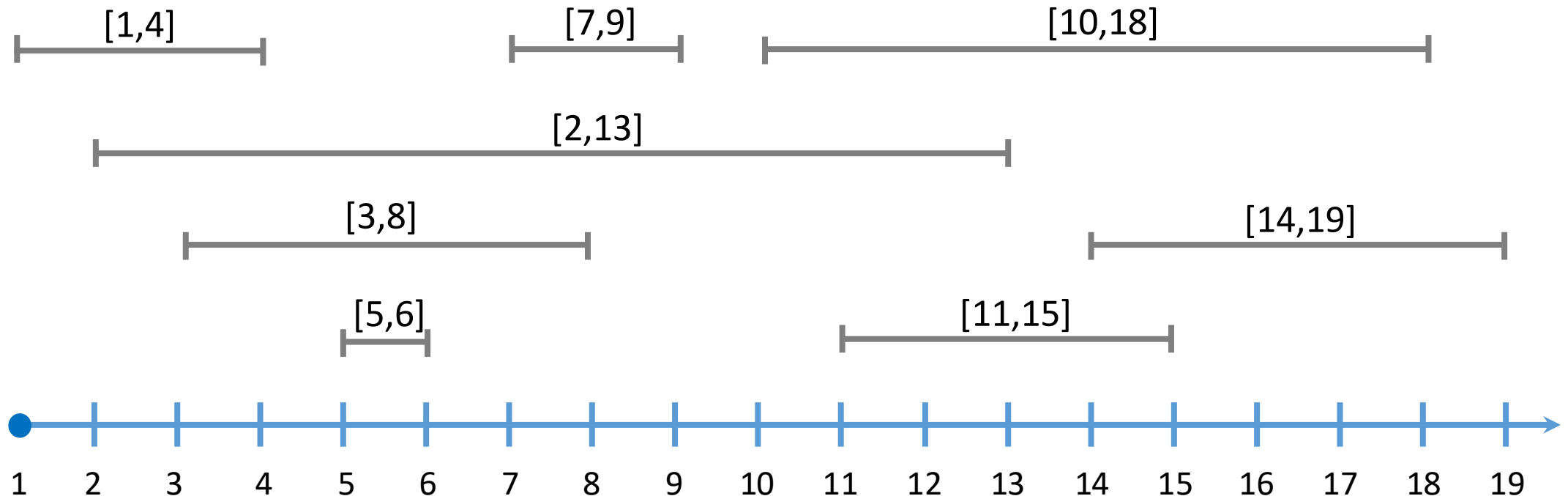
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Algorithm 3

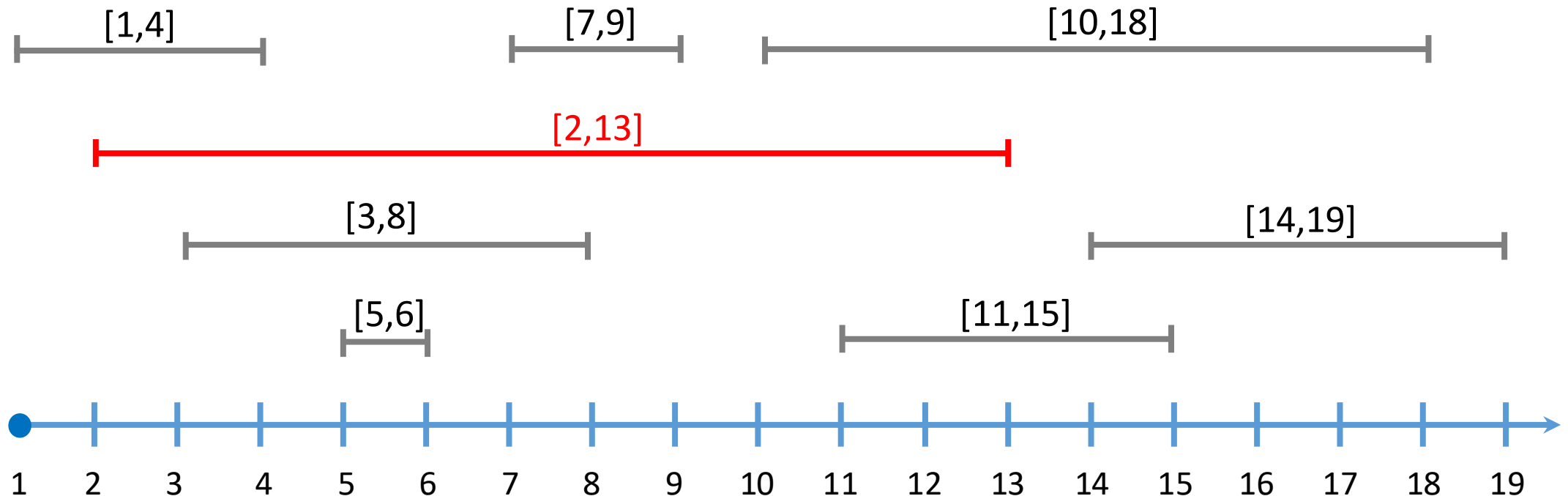
Algorithm 3

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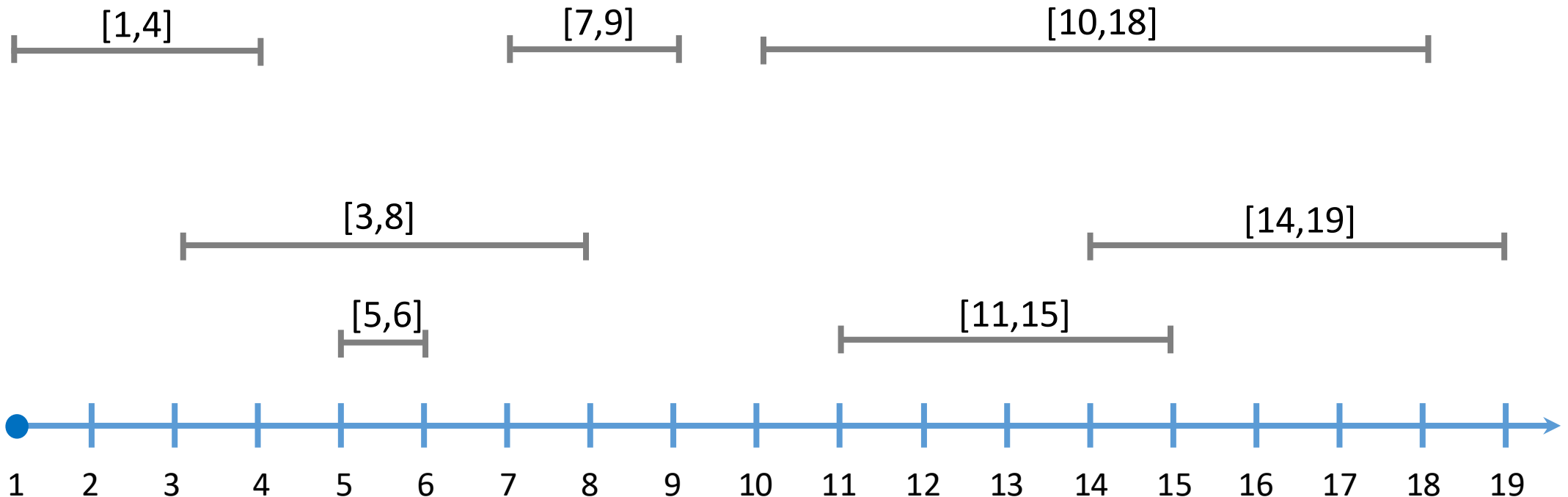
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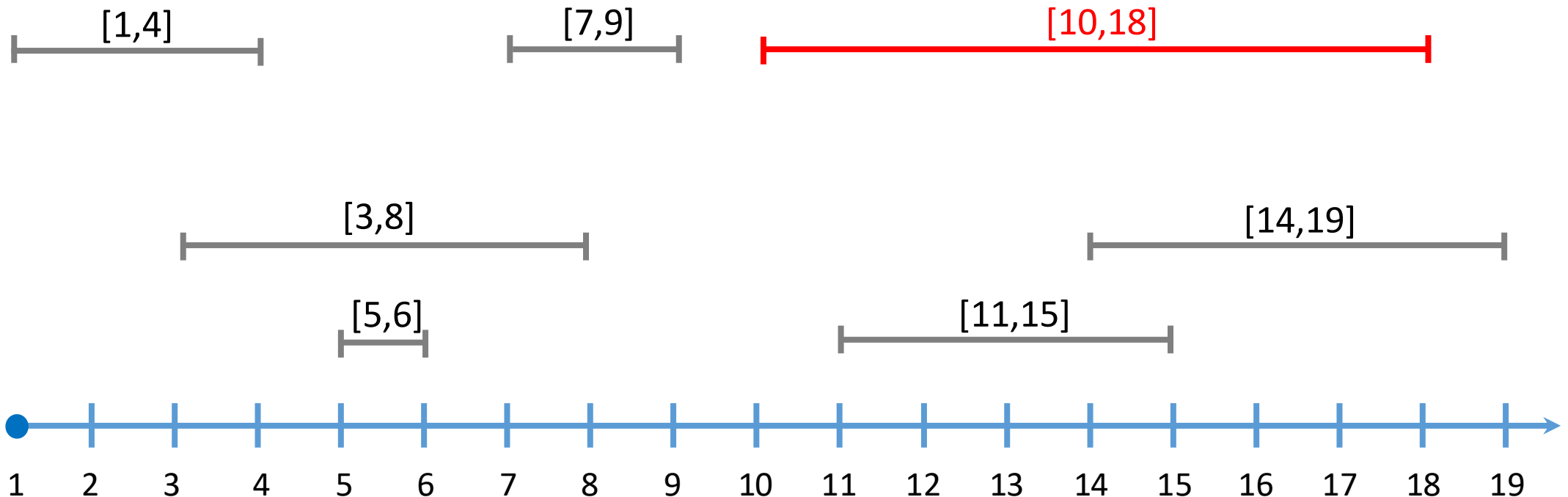
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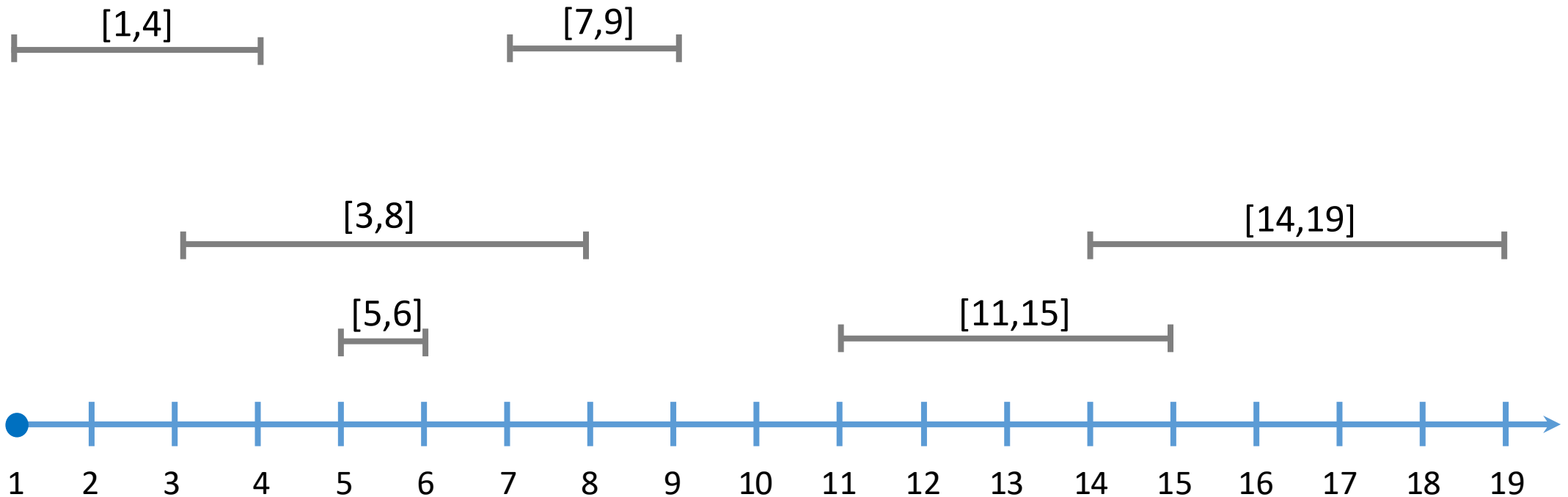
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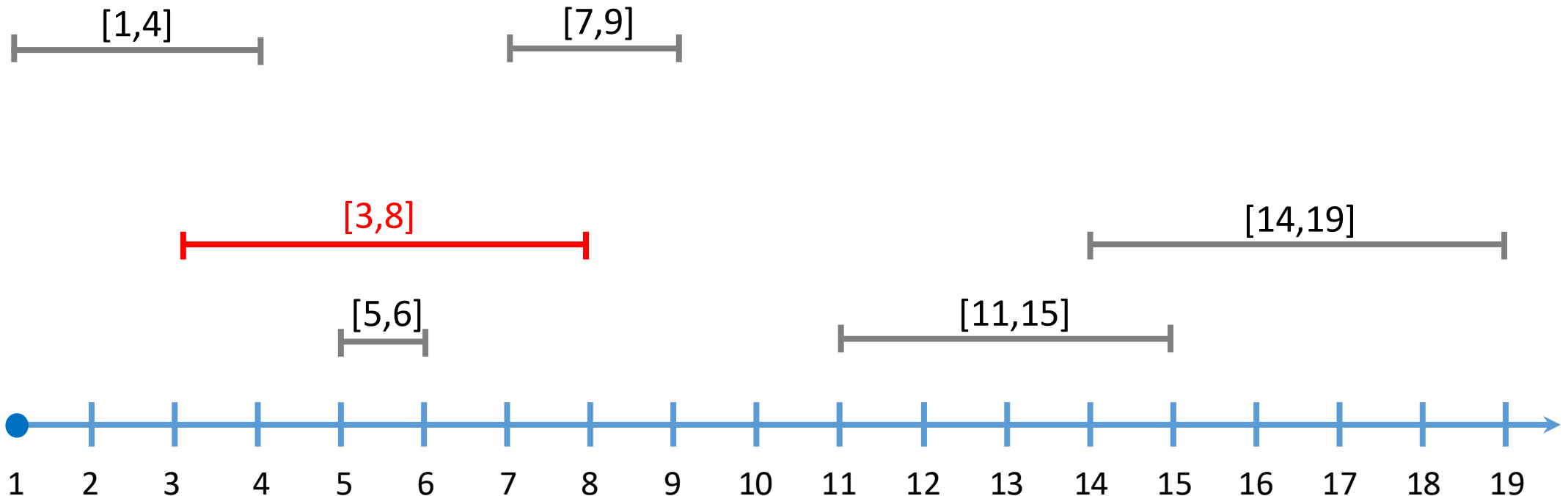
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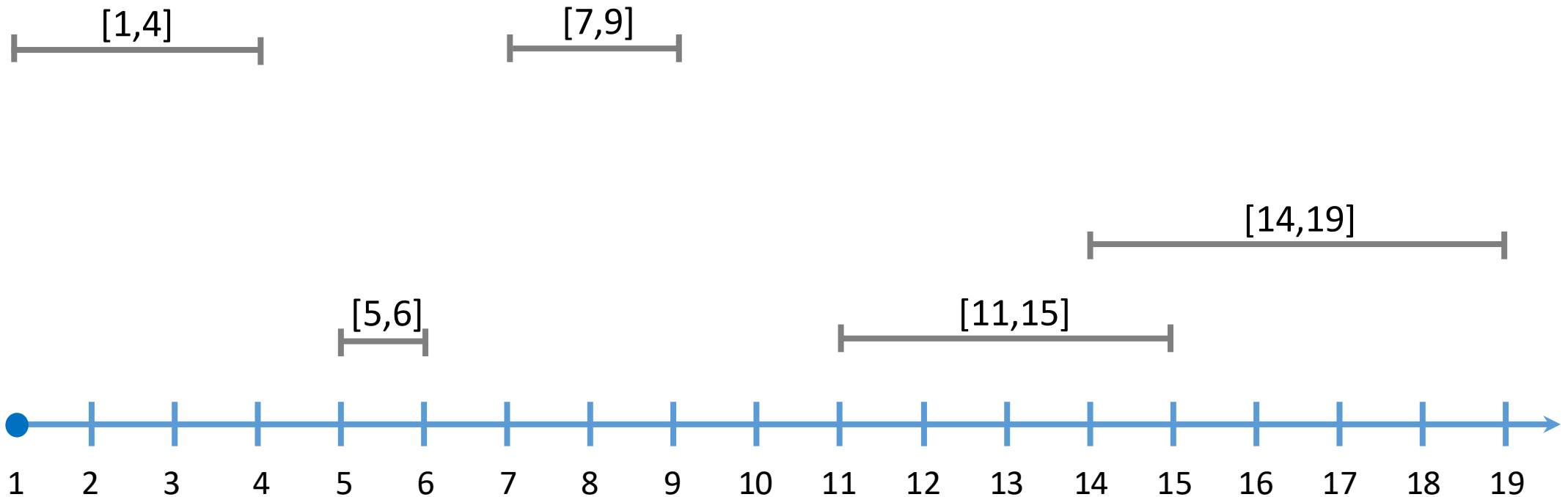
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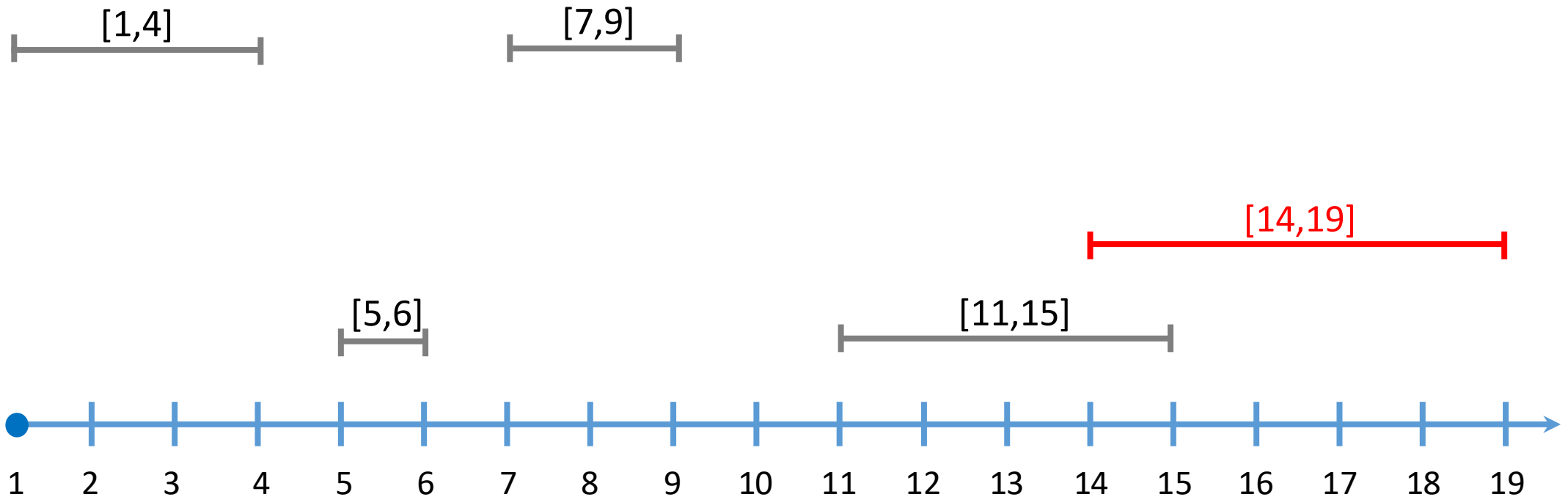
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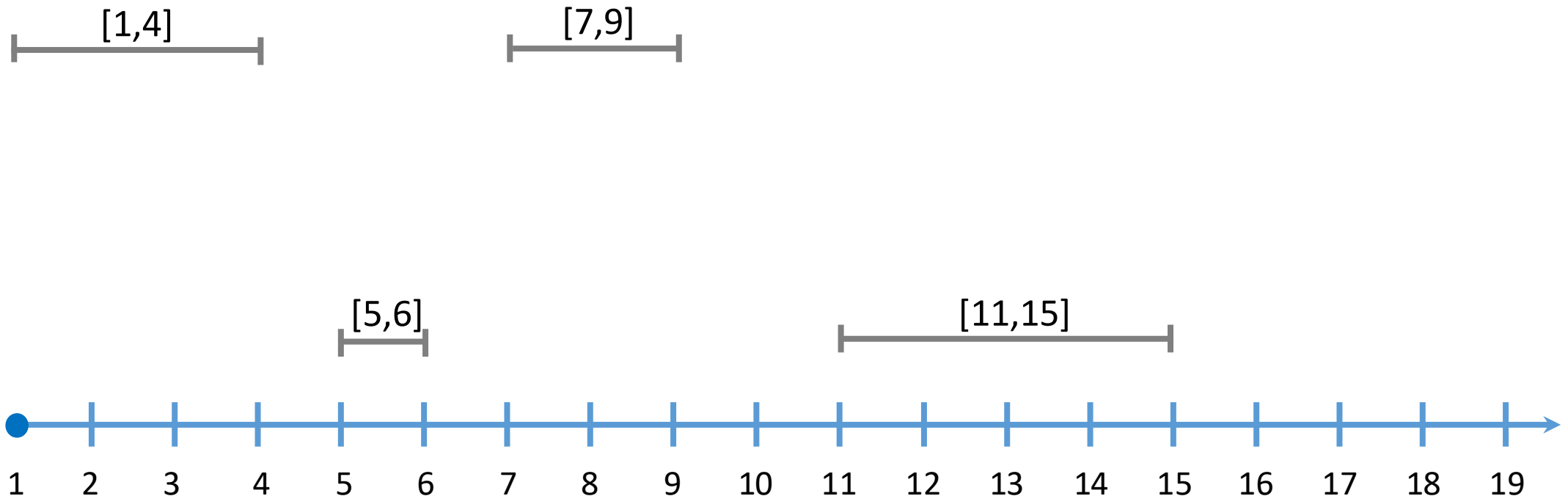
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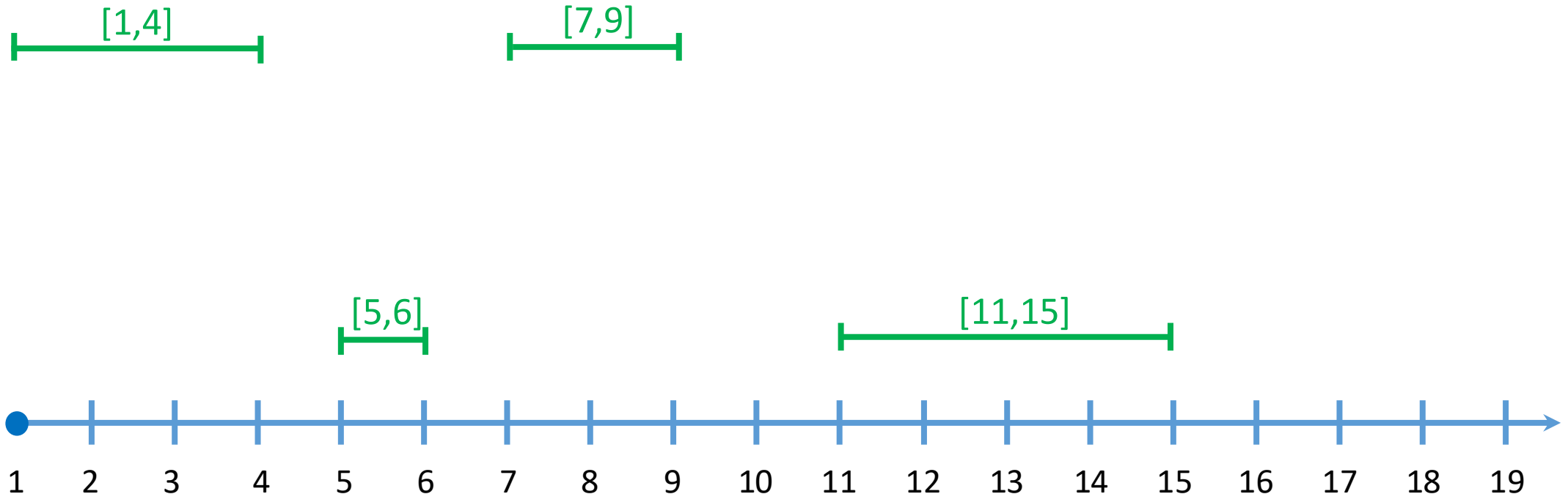
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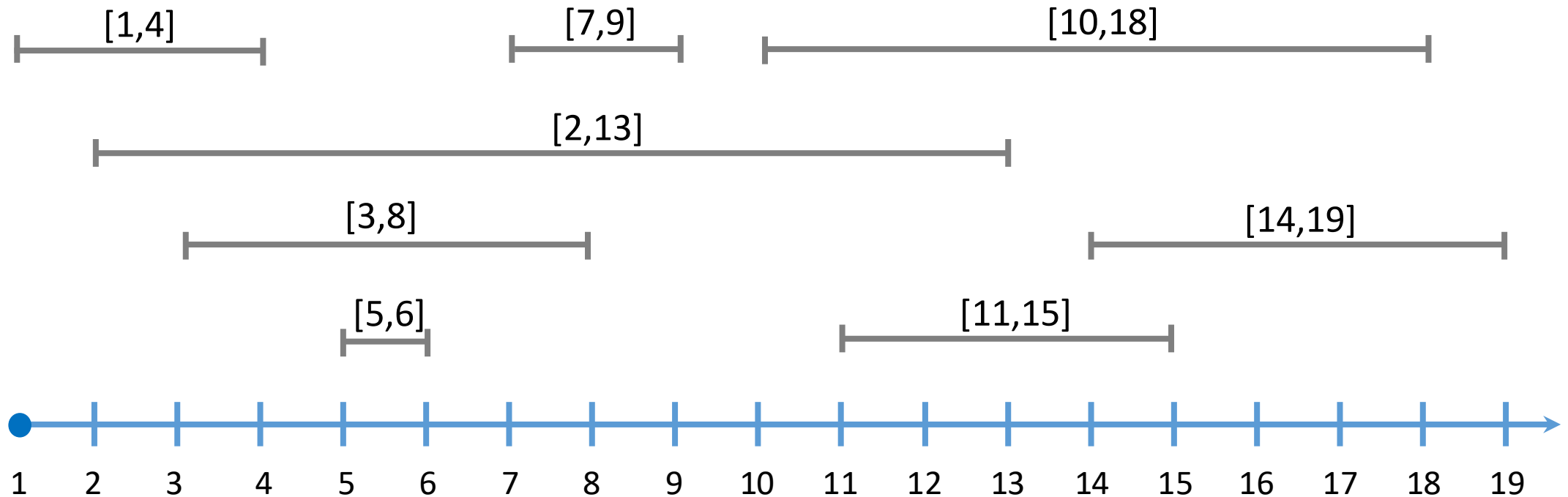
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Algorithm 4

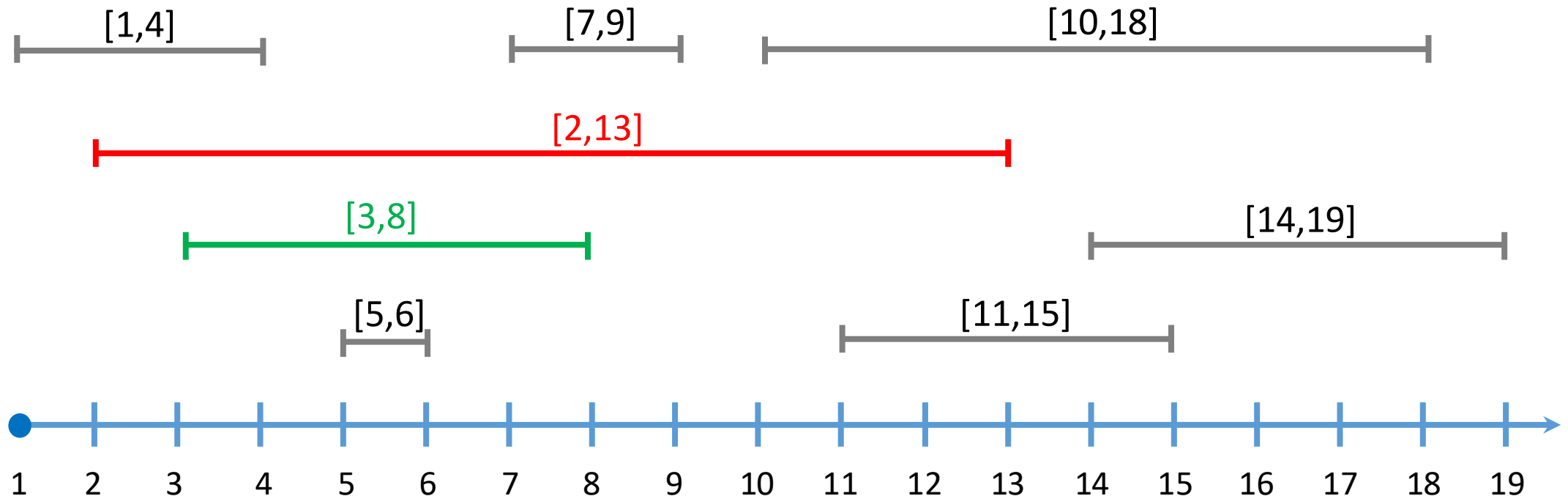
Algorithm 4

- Let us propose an algorithm: DELETE-DOMINATING-INTERVALS.
- From left to right, delete all intervals that dominate another interval.



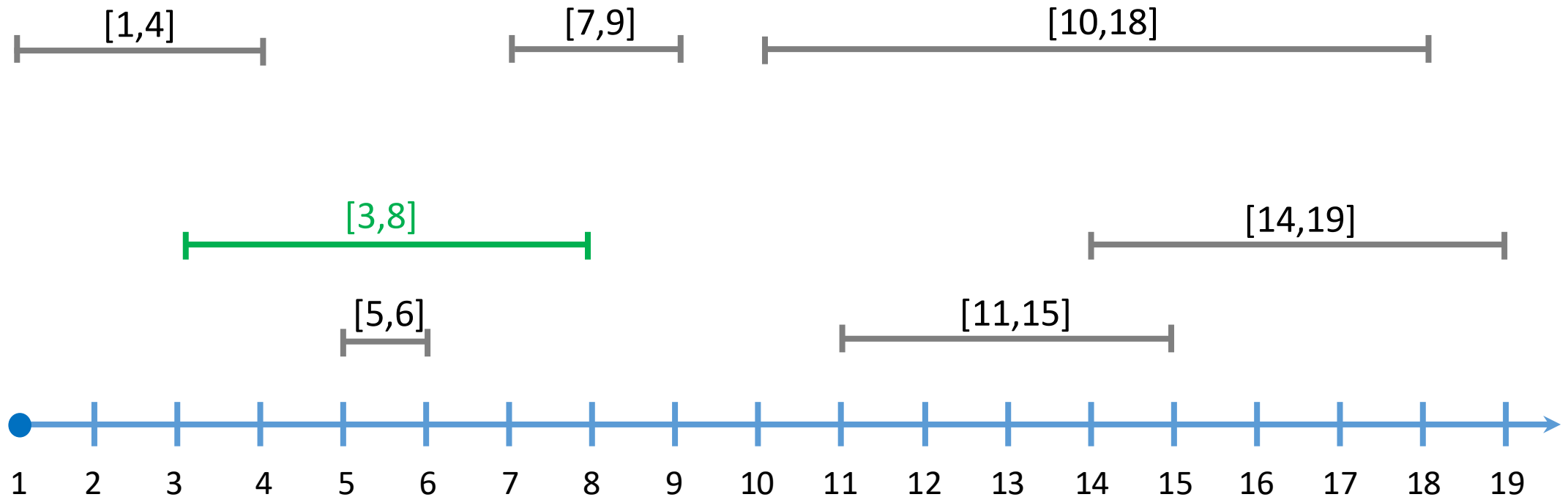
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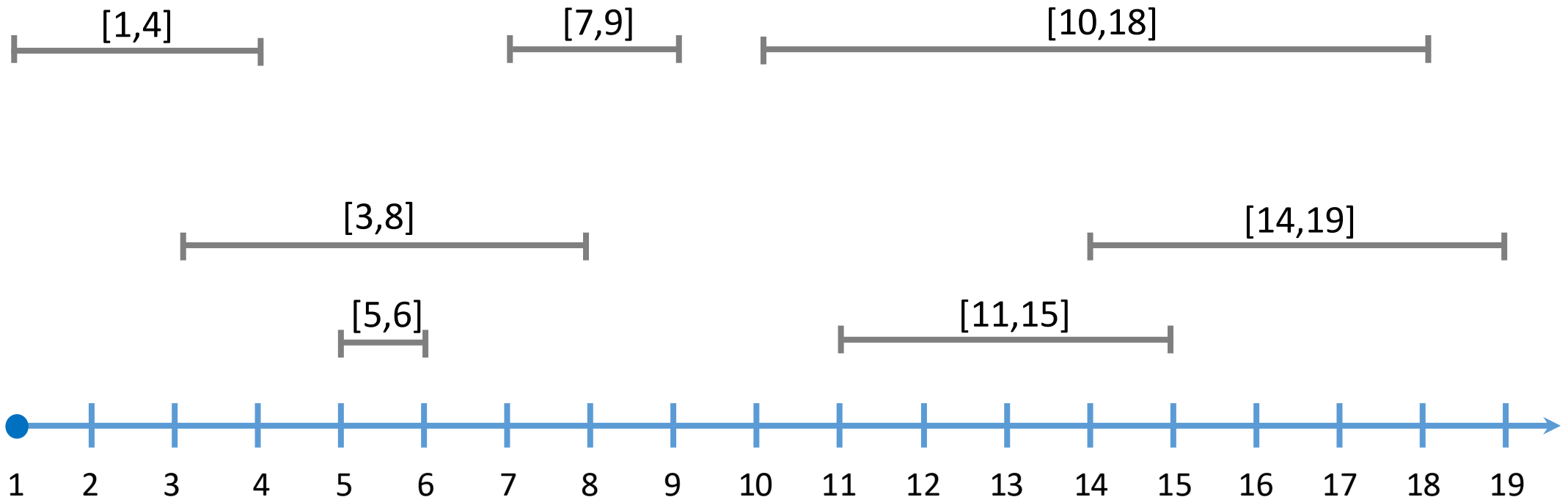
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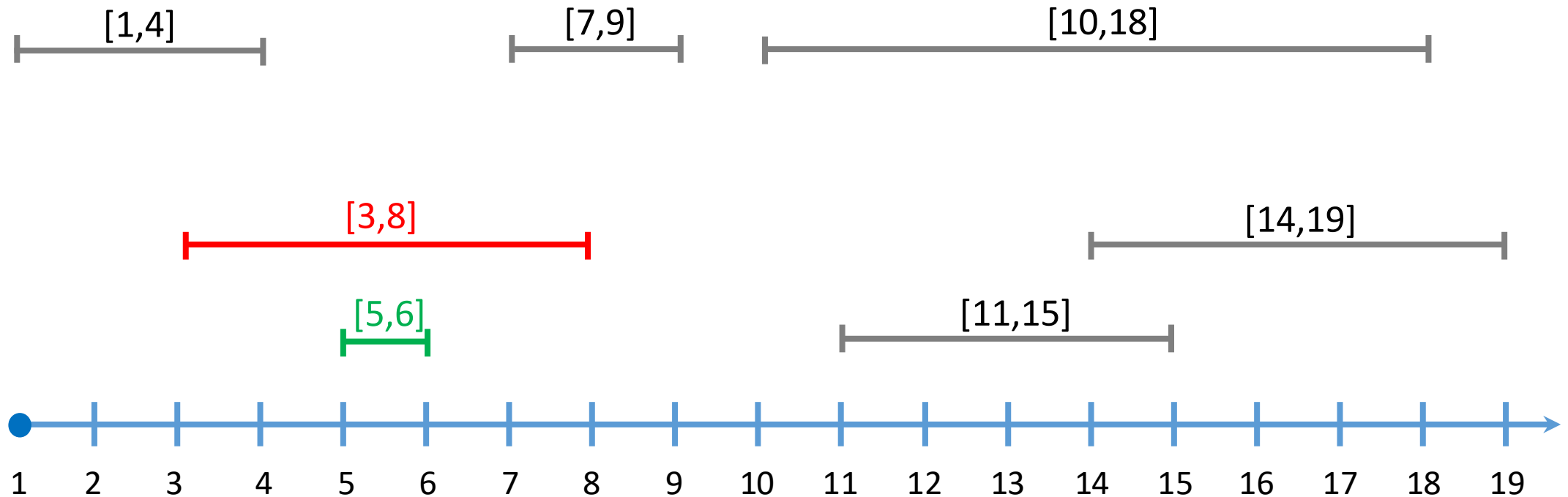
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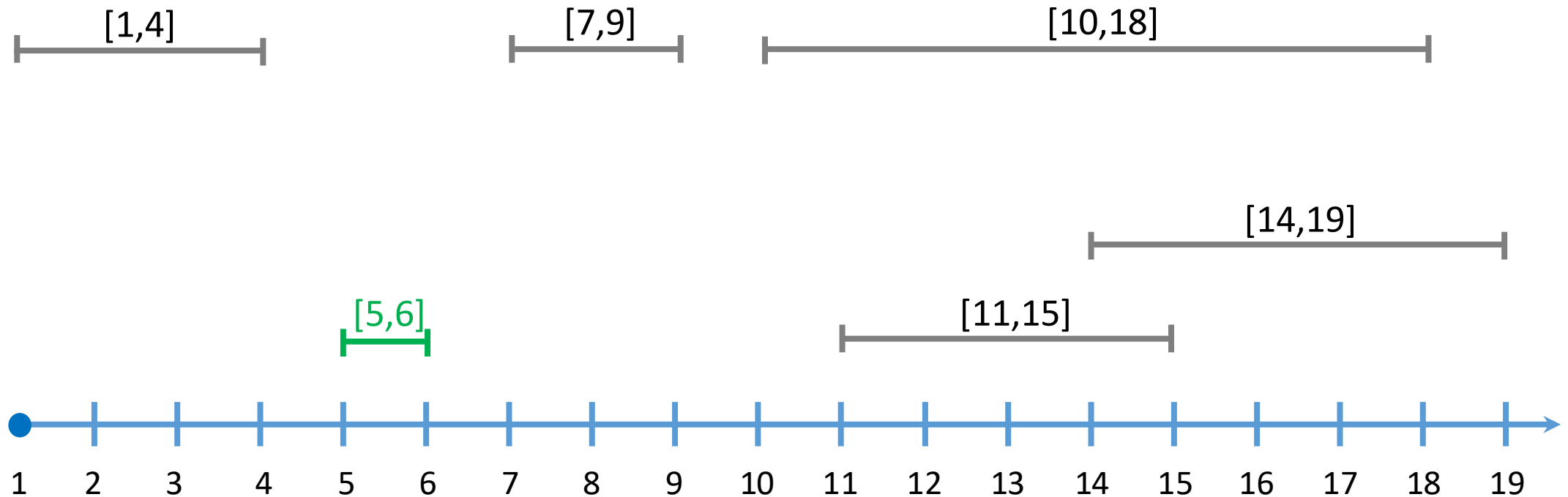
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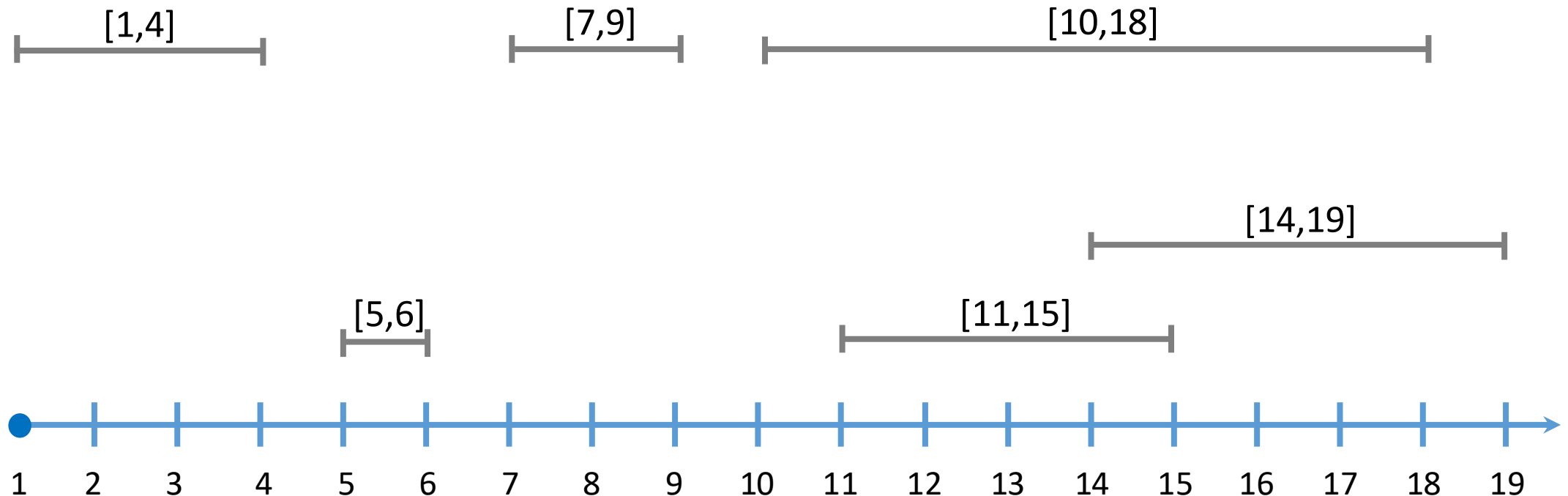
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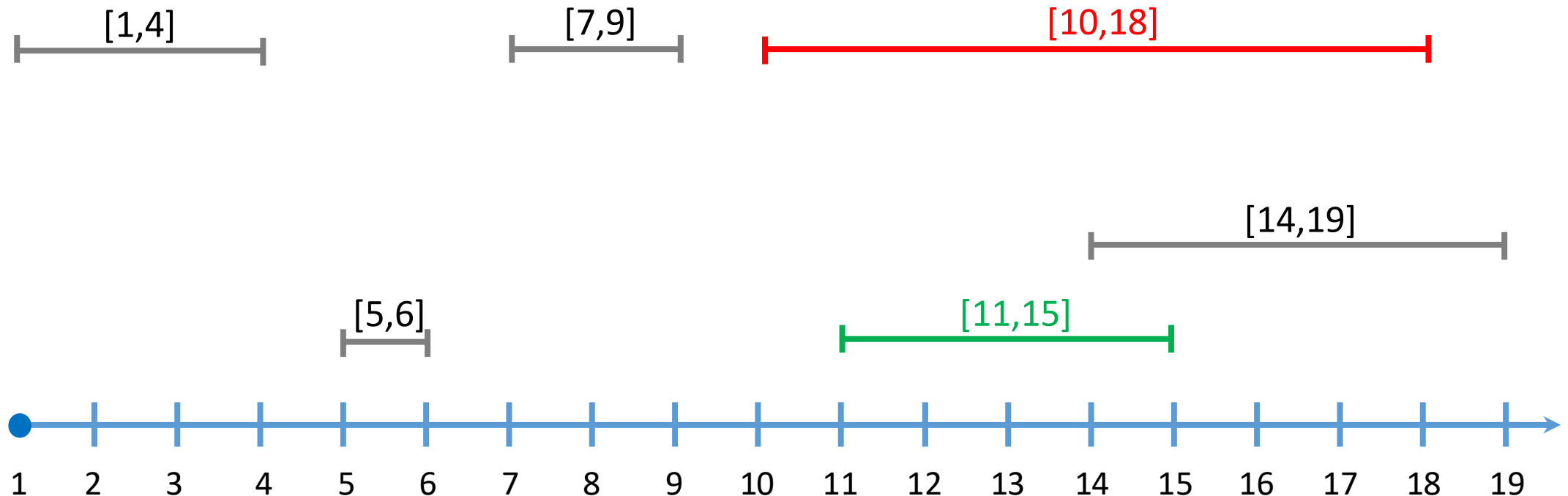
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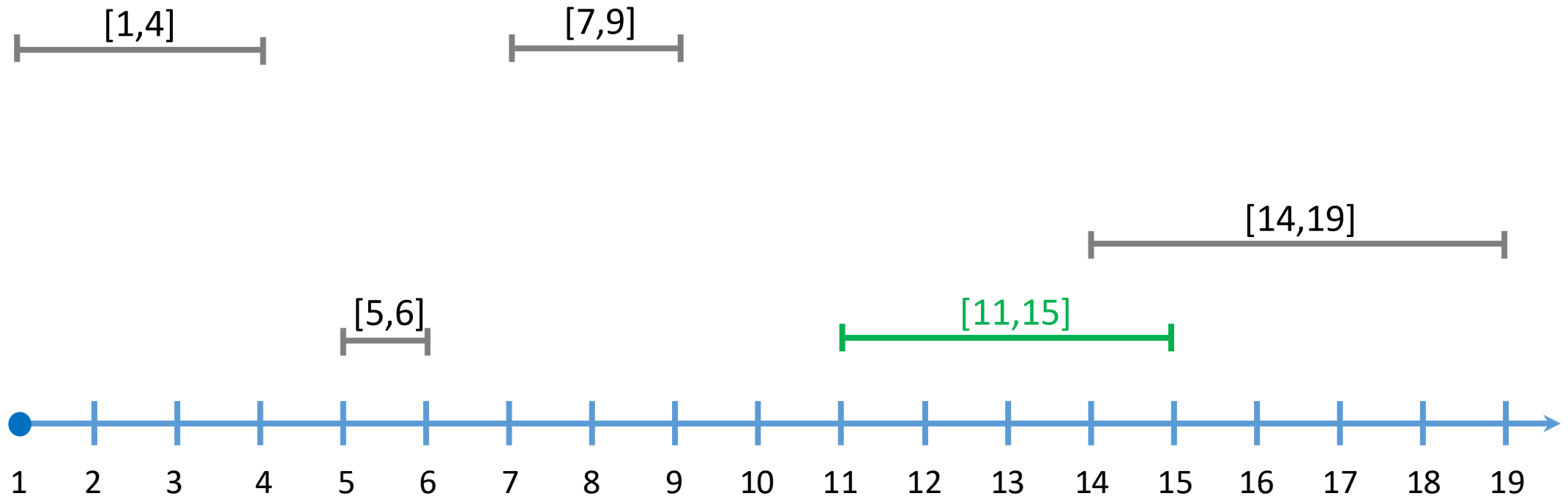
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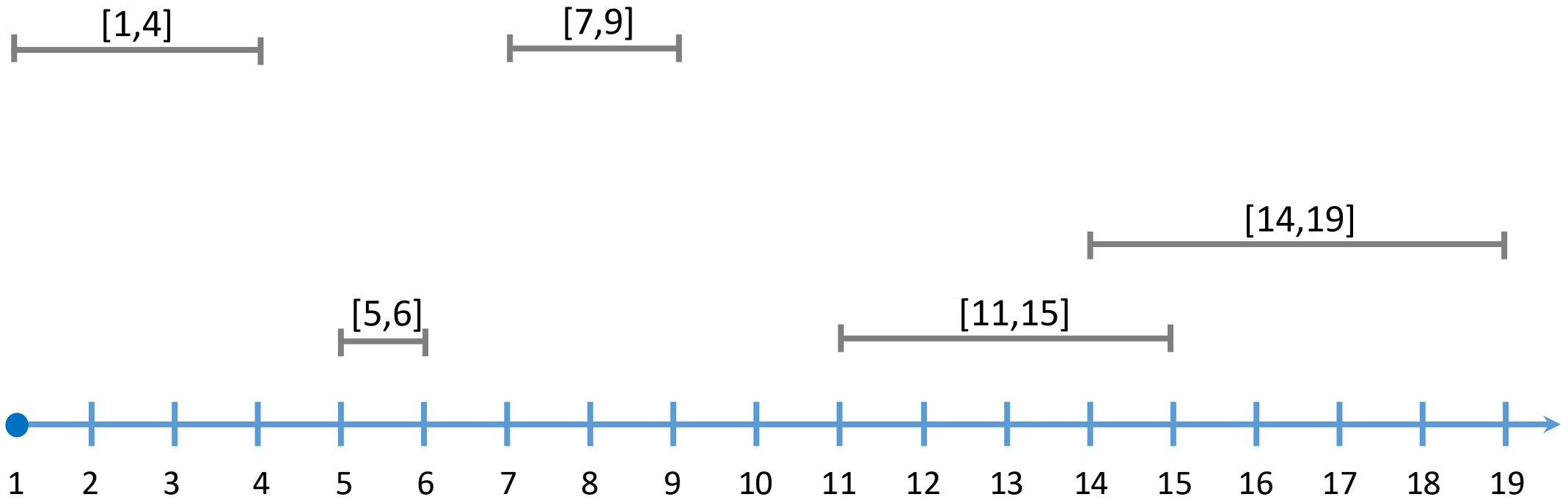
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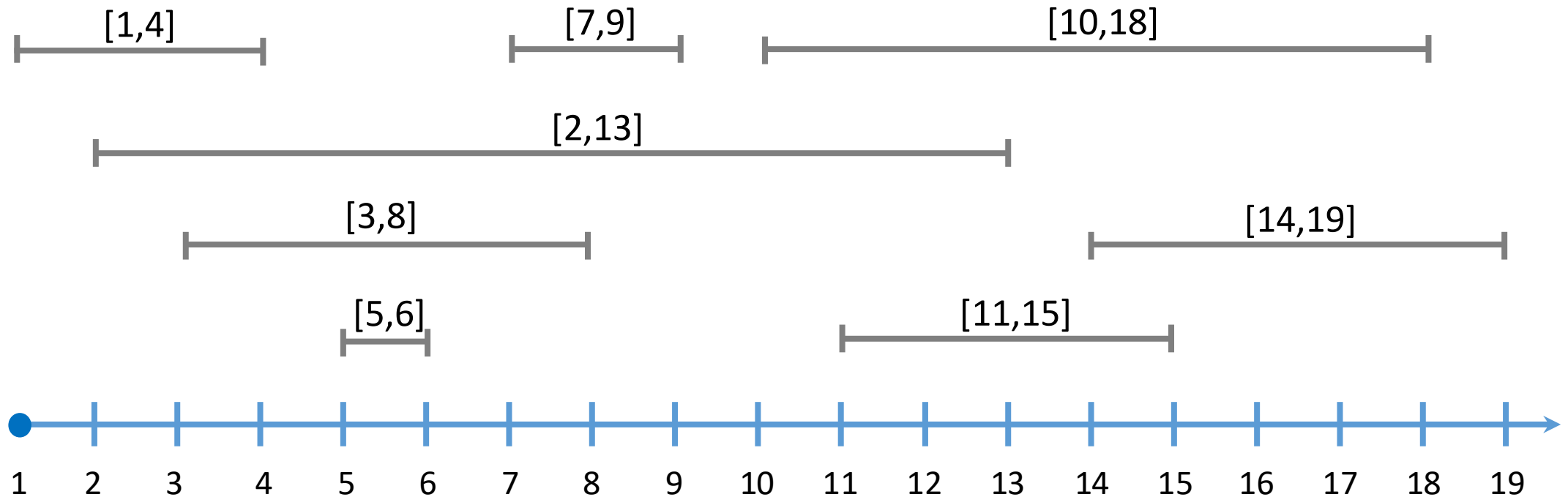
Algorithm 4

- Does **Algorithm 2** always output a set of intervals that don't clash with each other?
- NO.

Algorithm 5

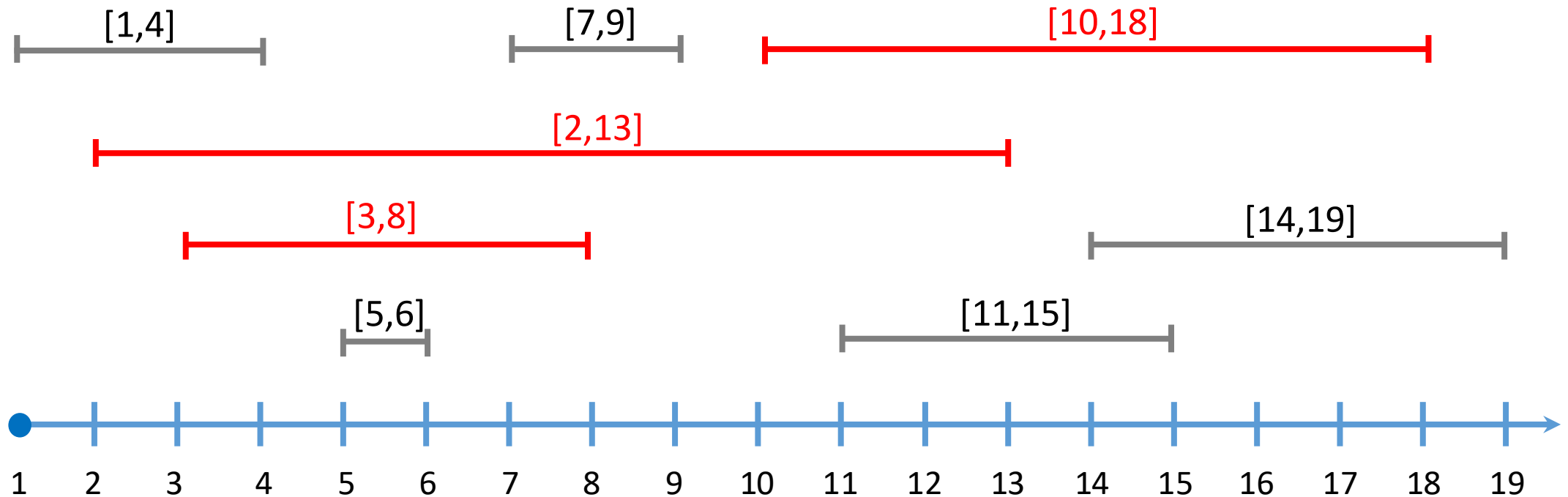
Algorithm 5

- Algorithm 5 = Algorithm 4 + Algorithm 1.
 - First, DELETE-DOMINATING-INTERVALS.
 - Then, EARLIEST-START-TIME.
-



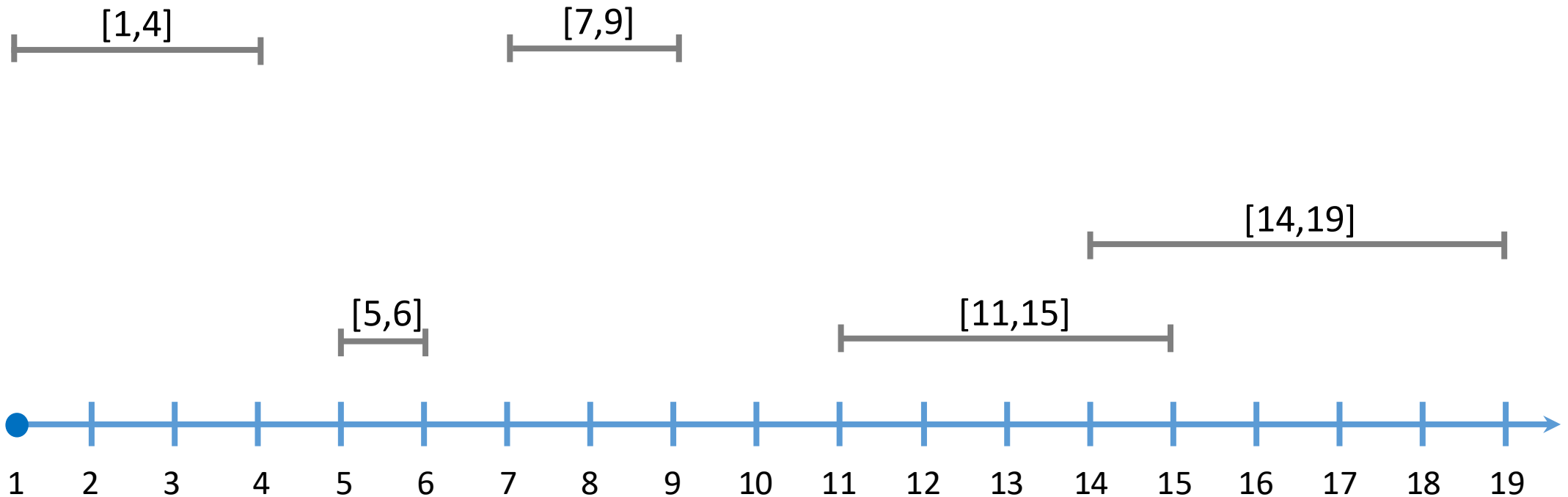
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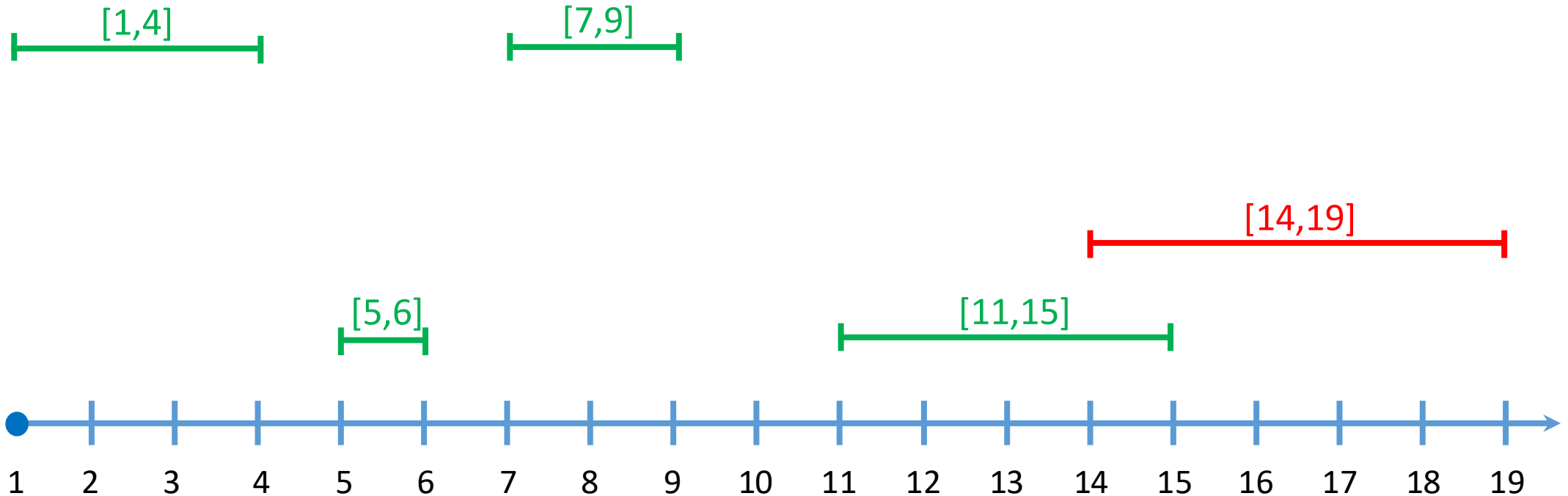
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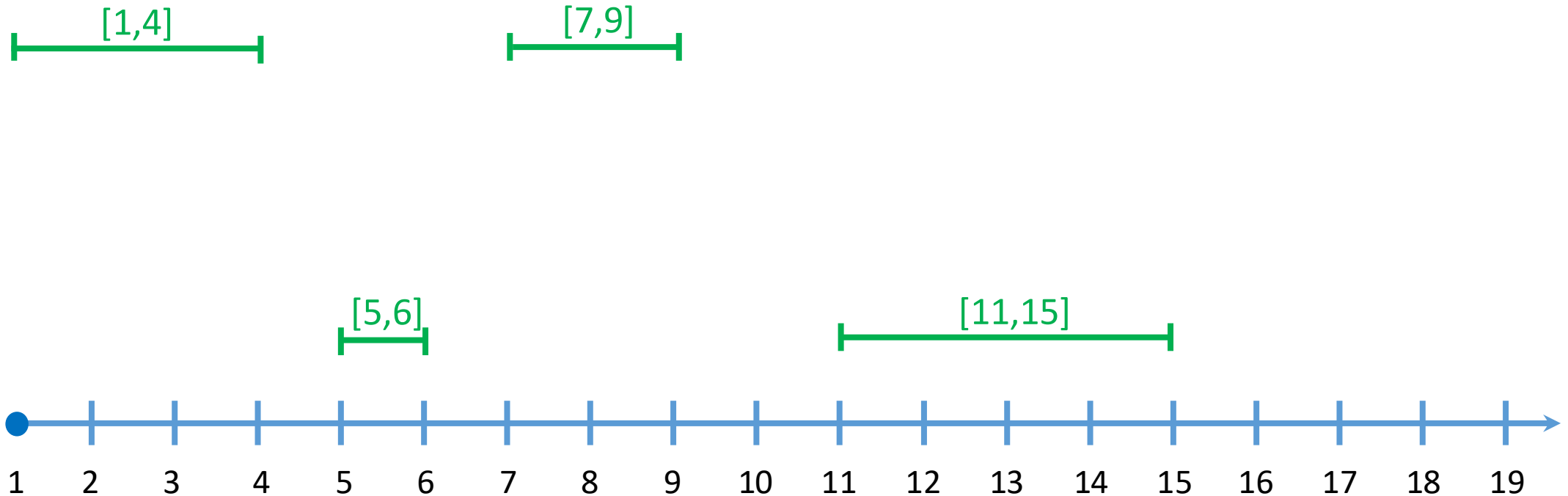
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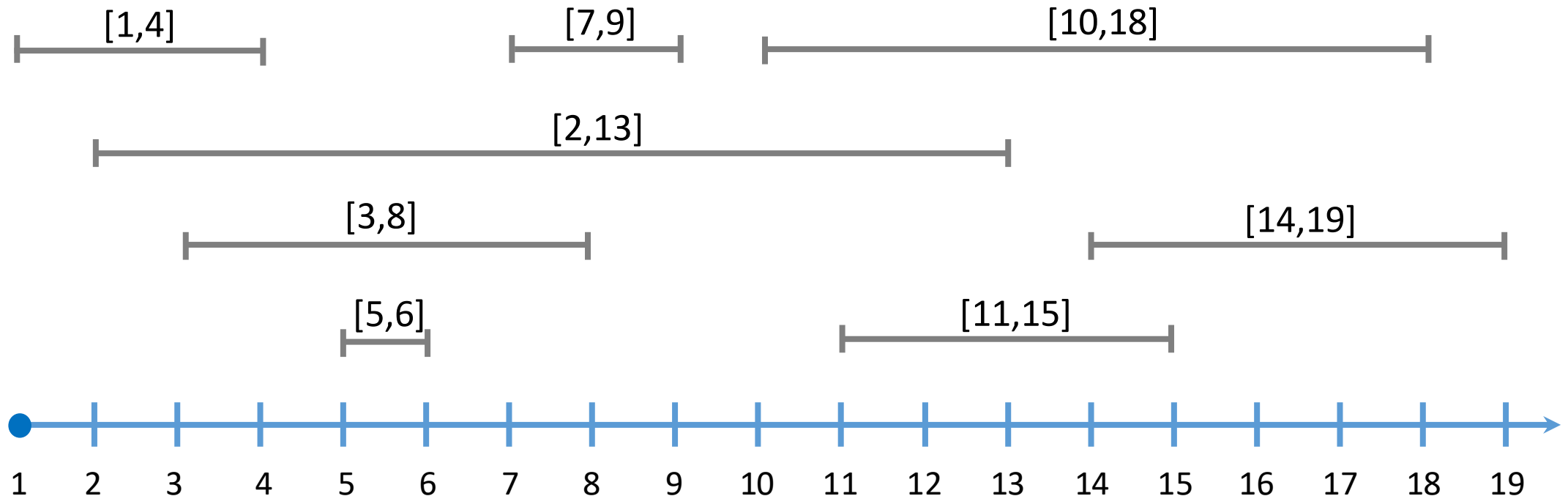
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Algorithm 6

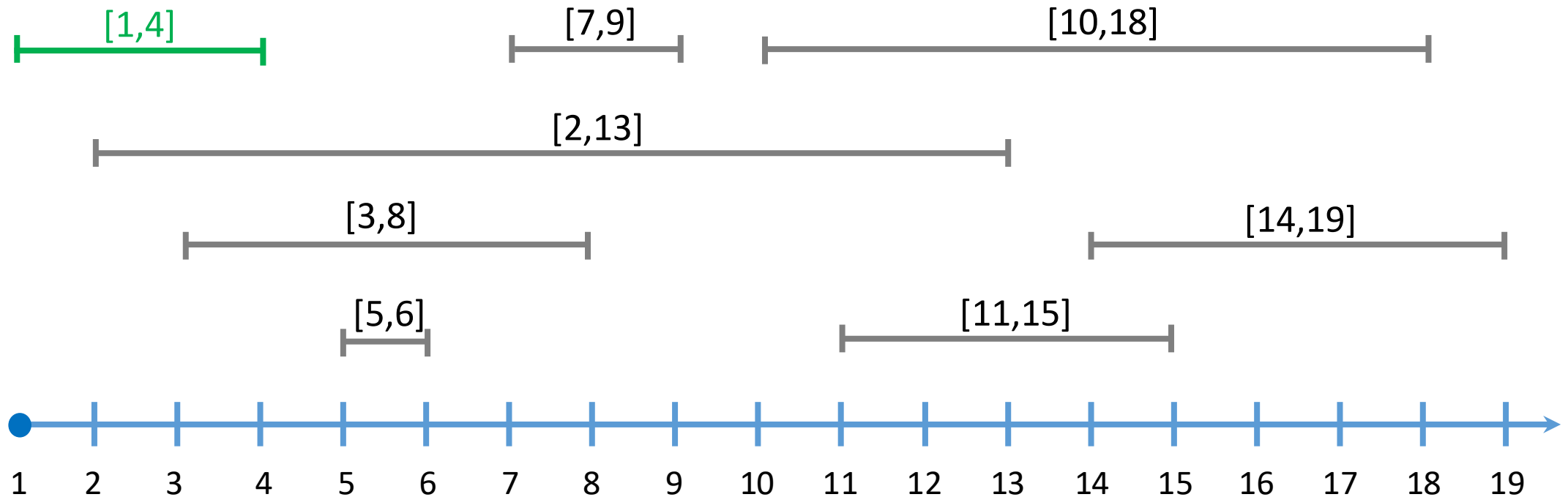
Algorithm 6

- Let us propose an algorithm: EARLIEST-END-TIME.
- From left to right, select the interval with the earliest end time.



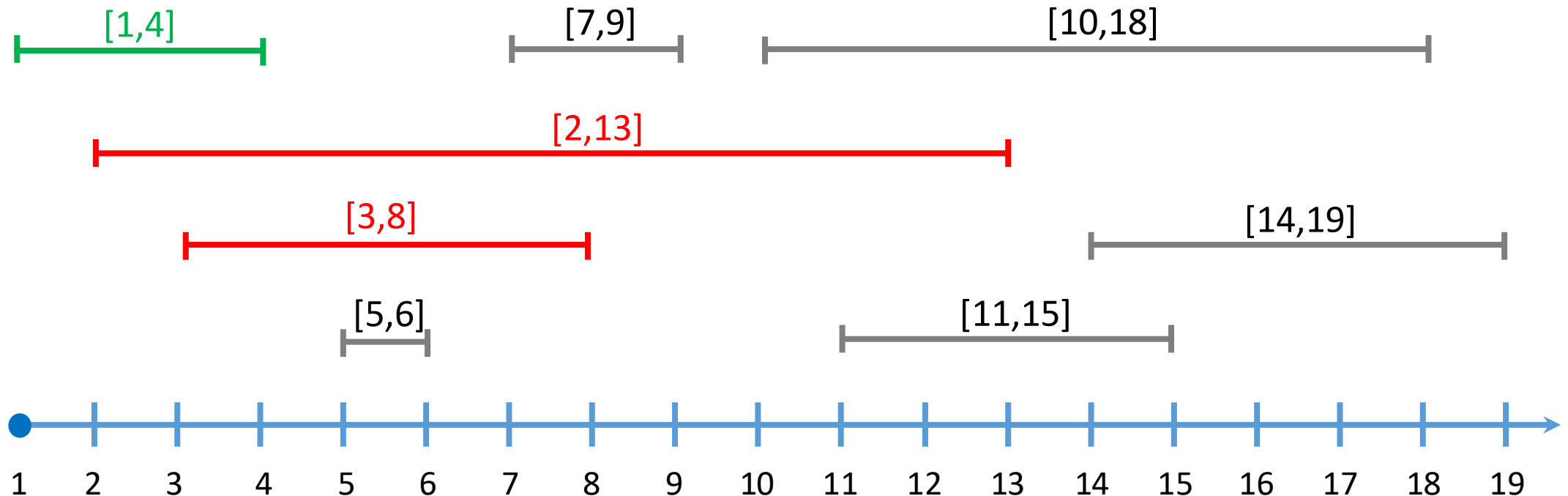
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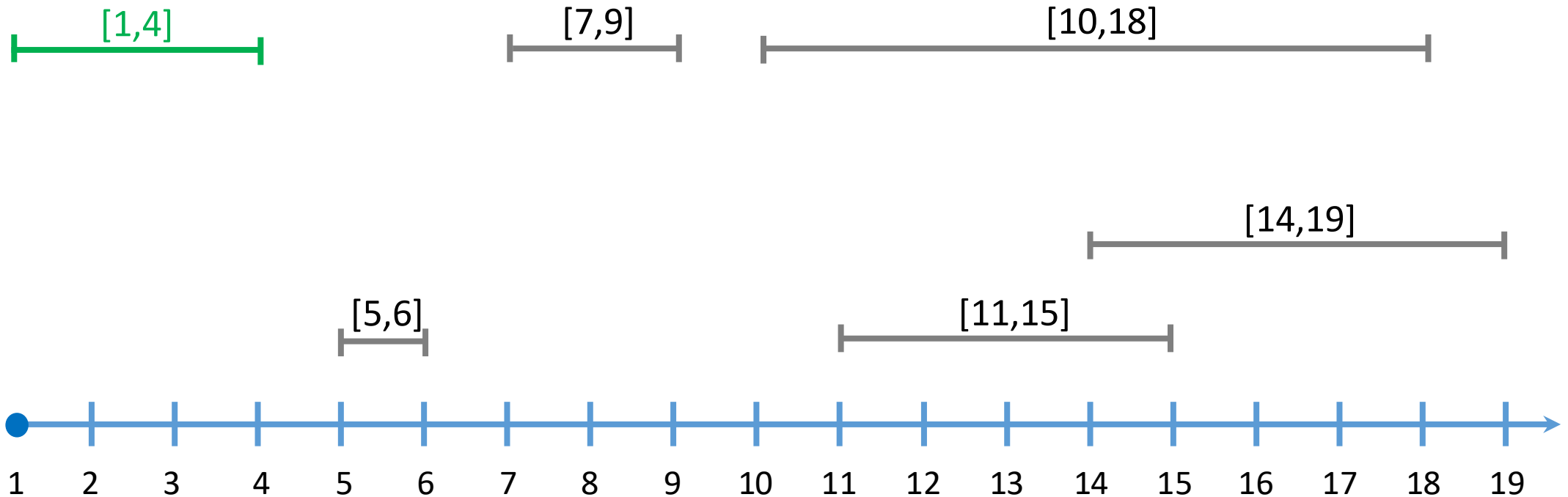
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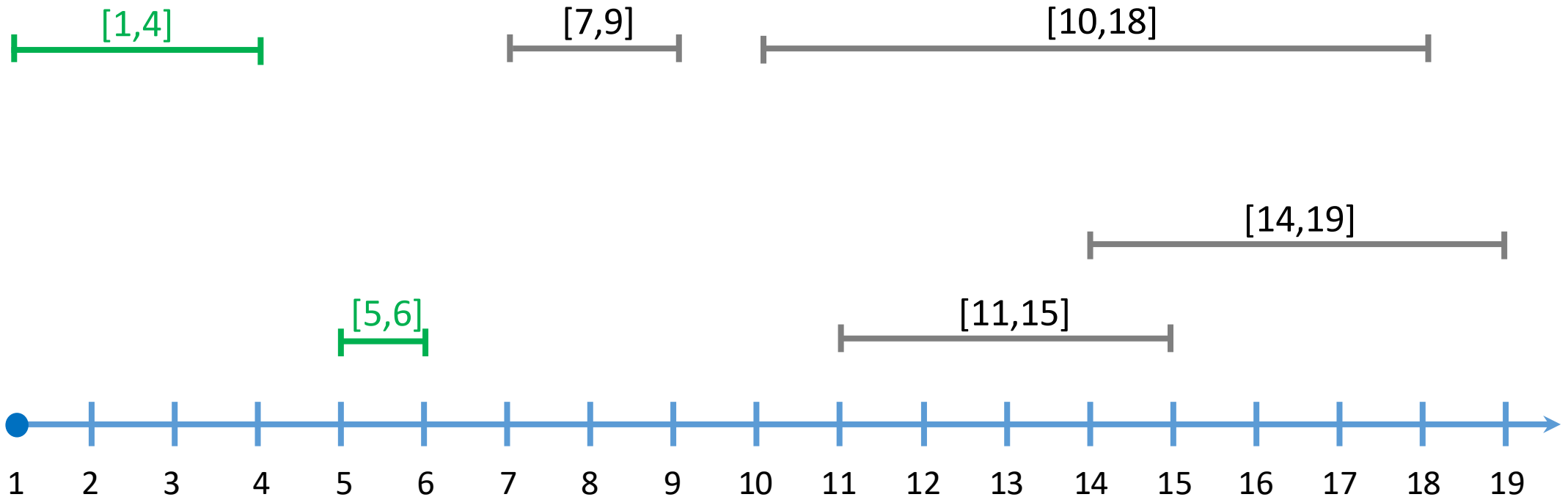
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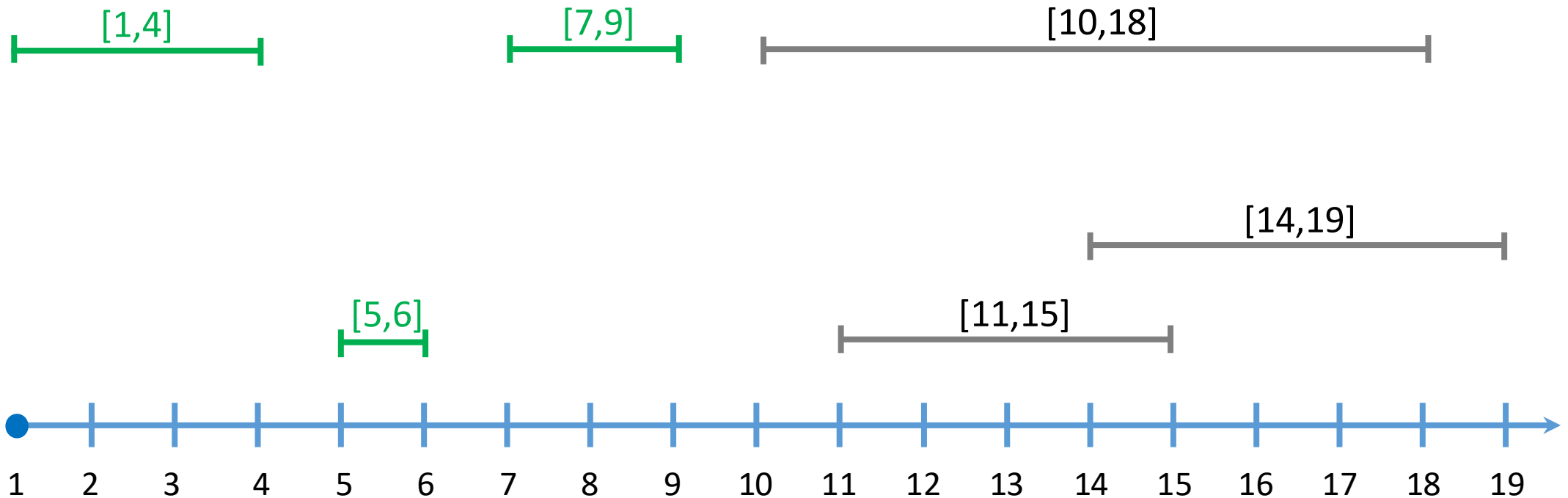
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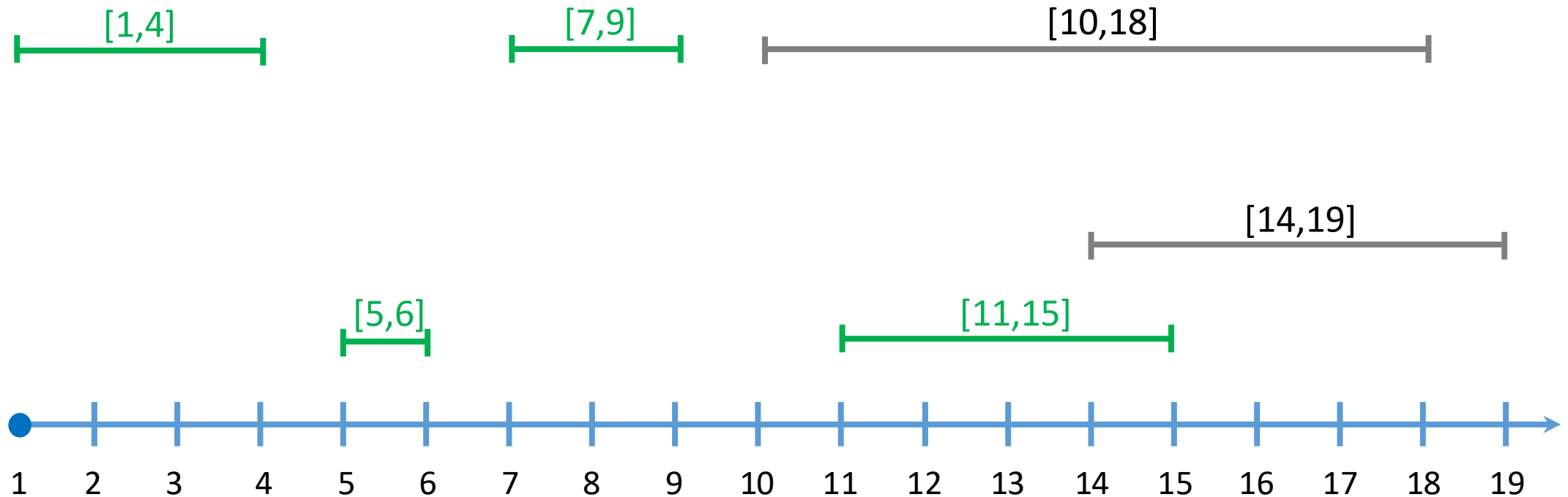
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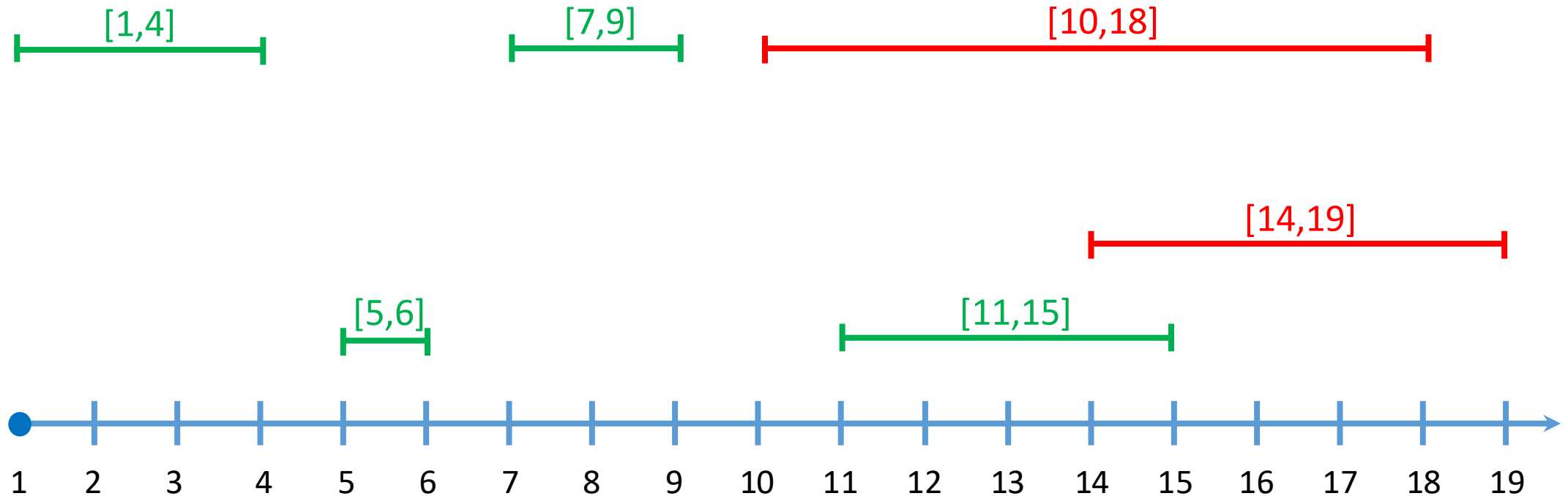
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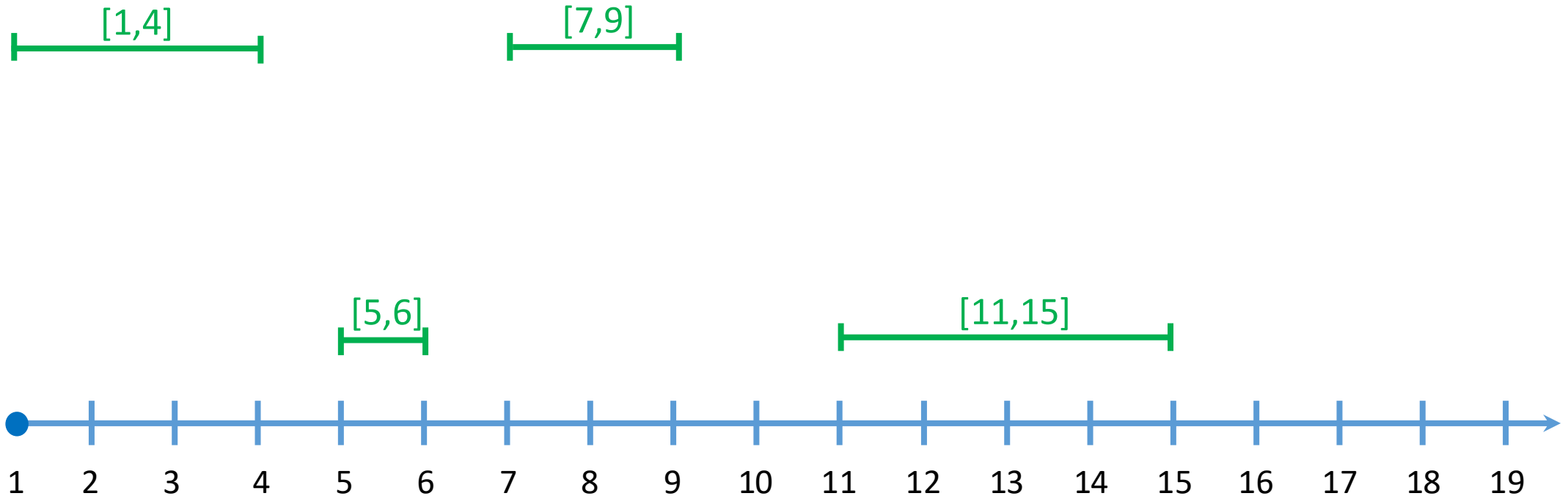
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Algorithm 6

- Does **Algorithm 2** always output a set of intervals that don't clash with each other?
- YES.
- Does **Algorithm 2** always output the maximum number of intervals that don't clash with each other?
- YES.

Algorithms for Interval Scheduling

Algorithm Number	Algorithm Name	Correct?	Optimal?
Algorithm 1	EARLIEST-START-TIME	YES	NO
Algorithm 2	SELECT-SMALLEST-INTERVAL	YES	NO
Algorithm 3	DELETE-LARGEST-INTERVAL	YES	NO
Algorithm 4	DELETE-DOMINATING-INTERVALS	NO	-
Algorithm 5	Algorithm 4 + Algorithm 1	YES	YES
Algorithm 6	EARLIEST-END-TIME	YES	YES

Other Applications of Interval Scheduling

- Scheduling jobs (tasks) at a server
- Connecting flight planning at an airport
- Cargo shipping transfers at a seaport
- Etymology (origin/history) of words
- Genealogy (family tree) of species

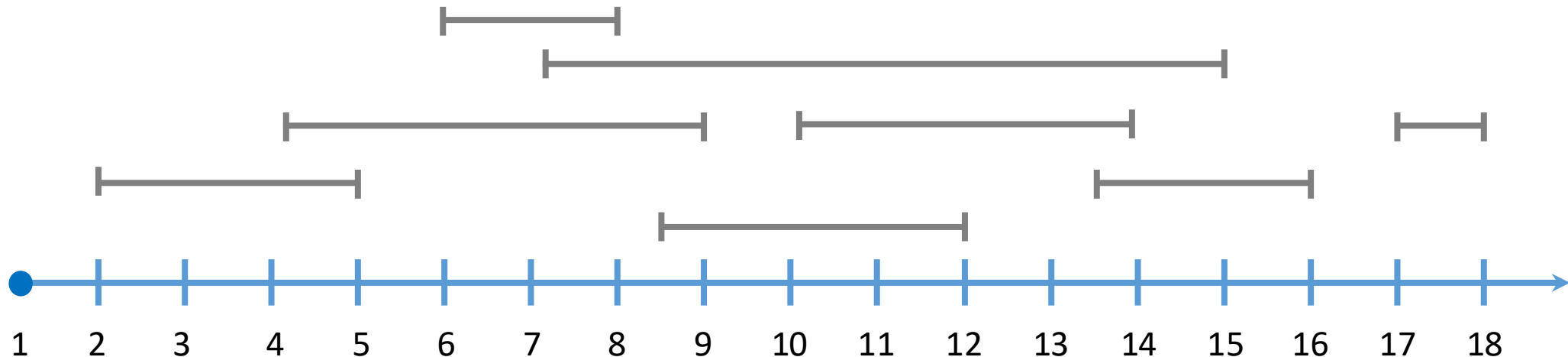
Time period: seconds/minutes

Time period: minutes/hours

Time period: days/weeks

Time period: hundreds of years

Time period: thousands of years

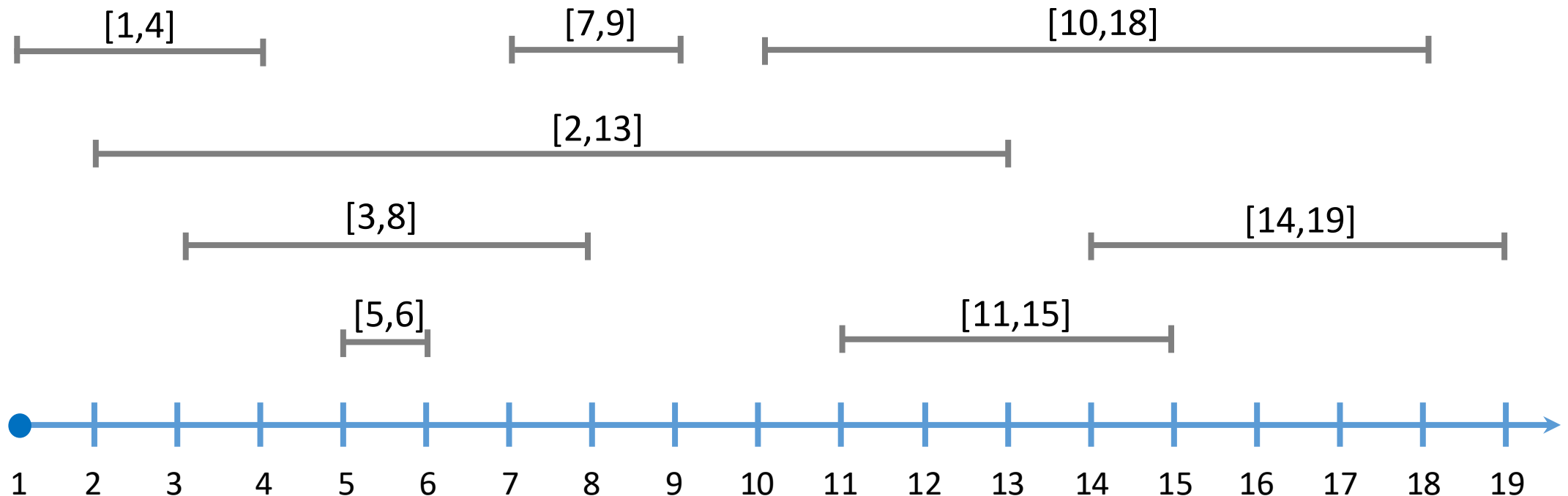


Thank You!



Homework

- The time period for each class is given by its instructor.
 - What is the minimum number of classrooms required to schedule the classes so that there are no clashes?
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